

WV_Thesis_Regression

Wes Vorley

2026-04-05

Read call data, remove N/A values and separate by date call was listed

```
calls <- read.csv("SPX_ATM_calls_MayJul2024.csv") %>%
  na.omit() %>%
  mutate(
    quote_date = as.Date(date), .before = expiration_date,
    expiration_date = as.Date(expiration_date)
  ) %>%
  select(-date)
```

Obtain daily log returns of SPX from May 2023 to August 2025, add column for 21-trading-day rolling volatility, and annualized using $\sqrt{252}$.

```
spx <- tq_get("^GSPC", get = "stock.prices",
  from = "2024-03-01",
  to = "2024-08-01")

returns <- spx %>%
  tq_transmute(select = adjusted,
    mutate_fun = periodReturn,
    period = "daily",
    col_rename = "daily_return") %>%
  mutate(log_return = log1p(daily_return)) %>%
  mutate(
    roll_sd = slide_dbl(log_return, sd, .before = 21, .complete = TRUE),
    mth_vol = roll_sd * sqrt(252)
  ) %>%
  as_tibble()
```

Function to filter the previous 21 trading days

```
hist_vol <- function(obs_date, lookback = 21) {
  obs_date <- as.Date(obs_date)

  returns %>%
    filter(date <= obs_date) %>%
    slice_tail(n = lookback) %>%
    mutate(
      x_day = as.numeric(date - obs_date),
      x = x_day / 252
    )
}
```

Function to fit linear, quadratic and exponential models to data

```
fit_vol <- function(vol_dat) {  
  const <- vol_dat$mth_vol[nrow(vol_dat)]  
  
  lin <- lm(mth_vol ~ x, data = vol_dat)  
  
  quad <- lm(mth_vol ~ x + I(x^2), data = vol_dat)  
  
  # Use log-linear model to get good starting values  
  log_fit <- lm(log(mth_vol) ~ x, data = vol_dat)  
  start_a <- exp(coef(log_fit)[1])  
  start_b <- coef(log_fit)[2]  
  exp <- nlsLM(  
    mth_vol ~ a * exp(b * x),  
    data = vol_dat,  
    start = list(a = start_a, b = start_b)  
  )  
  
  # Return data AND models together in a list  
  list(  
    vol_dat = vol_dat,  
    const = const,  
    lin = lin,  
    quad = quad,  
    exp = exp  
  )  
}
```

Functions for integrated variance of each model

```

V_lin <- function(t, fit) {
  # sigma(t) = a*t + b
  a <- coef(fit)[2] # slope
  b <- coef(fit)[1] # intercept

  res <- (1/3) * a^2 * t^3 + (a * b) * t^2 + b^2 * t
}

V_quad <- function(t, fit) {
  # sigma(t) = a*t^2 + b*t + c
  a <- coef(fit)[3]
  b <- coef(fit)[2]
  c <- coef(fit)[1]

  (1/5) * a^2 * t^5 +
  (1/2) * a * b * t^4 +
  ((2/3) * a * c + (1/3) * b^2) * t^3 +
  (b * c) * t^2 +
  c^2 * t
}

V_exp <- function(t, fit) {
  # sigma(t) = a*exp(b*t)
  a <- coef(fit)[1]
  b <- coef(fit)[2]

  ifelse(
    abs(b) < 1e-8,
    a^2 * t,
    (a^2 / (2 * b)) * (exp(2 * b * t) - 1)
  )
}

```

Functions for Black-Scholes model with constant and deterministic volatility

```

bs_const_vol <- function(S, K, r, t, vol) {
  d1 <- (log(S / K) + (r + 0.5 * vol^2) * t) / (vol * sqrt(t))
  d2 <- d1 - vol * sqrt(t)

  S * pnorm(d1) - K * exp(-r * t) * pnorm(d2)
}

bs_det_vol <- function(S, K, r, t, V) {
  V <- pmax(V, 1e-10)
  d1 <- (log(S / K) + r * t + 0.5 * V) / sqrt(V)
  d2 <- d1 - sqrt(V)

  S * pnorm(d1) - K * exp(-r * t) * pnorm(d2)
}

```

Function to create a results table for call prices on each date

```

call_tbl_fun <- function(obs_date, vol_dat) {
  call_dat <- calls %>%
    filter(quote_date == obs_date) %>%
    mutate(
      x_day = as.numeric(as.Date(expiration_date) - as.Date(obs_date)),
      x      = x_day / 252,
    ) %>%
    relocate(x_day, .after = expiration_date) %>%
    relocate(x, .after = x_day)

  calls_tbl <- tibble(
    x_day      = call_dat$x_day,
    x          = call_dat$x,
    "Obs. Date" = call_dat$quote_date,
    "Exp. Date" = call_dat$expiration_date,
    "S"        = call_dat$spot_price,
    "K"        = call_dat$strike_price,
    "t"        = call_dat$maturity_yrs,
    "r"        = call_dat$interest_rate / 100,
    "True Call" = call_dat$close,

    "Const Call" = bs_const_vol(S, K, r, t, vol_dat$const),
    "Lin Call"   = bs_det_vol(S, K, r, t, V_lin(t, vol_dat$lin)),
    "Quad Call"  = bs_det_vol(S, K, r, t, V_quad(t, vol_dat$quad)),
    "Exp Call"   = bs_det_vol(S, K, r, t, V_exp(t, vol_dat$exp)),

    # Errors
    err_const = `True Call` - `Const Call`,
    err_lin   = `True Call` - `Lin Call`,
    err_quad  = `True Call` - `Quad Call`,
    err_exp   = `True Call` - `Exp Call`,

    abs_err_const = abs(err_const),
    abs_err_lin   = abs(err_lin),
    abs_err_quad  = abs(err_quad),
    abs_err_exp   = abs(err_exp)
  )

  return(calls_tbl)
}

```

Function to create a plot for 21-day rolling volatilities up to a given date

```

plot_vol <- function(obs_date) {
  # Build volatility table
  fit <- hist_vol(obs_date) %>% fit_vol()
  vol_dat <- fit$vol_dat

  # Build x-axis breaks
  x_min <- min(vol_dat$x_day)
  x_max <- max(vol_dat$x_day)
  x_breaks <- seq(x_min, x_max, length.out = 5)
  x_labels <- format(as.Date(obs_date) + round(x_breaks), "%b %d")

  # Get ending date for title
  end_date <- max(as.Date(vol_dat$date))

  ggplot(vol_dat, aes(x = x_day, y = mth_vol)) +
    geom_point(size = 1.5) +
    geom_line(aes(y = fit$const, color = "Constant"), linewidth = 0.8) +
    geom_line(aes(y = predict(fit$lin), color = "Linear"), linewidth = 0.8) +
    geom_line(aes(y = predict(fit$quad), color = "Quadratic"), linewidth = 0.8) +
    geom_line(aes(y = predict(fit$exp), color = "Exponential"), linewidth = 0.8) +
    scale_x_continuous(
      breaks = x_breaks,
      labels = x_labels
    ) +
    labs(
      # title = paste(
      #   "SPX 21-Trading-Day Rolling Volatility from",
      #   format(as.Date(obs_date), "%B %d"),
      #   "to",
      #   format(end_date, "%B %d, %Y")
      # ),
      x = "Date",
      y = "Monthly Volatility",
      color = "Model"
    ) +
    theme(plot.title = element_text(hjust = 0.5))
}

```

Output model performance for volatility fits on all dates

```

vol_errs_tbl <- function(obs_date) {
  fit      <- hist_vol(obs_date) %>% fit_vol()
  vol_dat <- fit$vol_dat

  vol_dat %>%
    transmute(
      `Obs. Date` = as.Date(obs_date),
      date,
      mth_vol,
      fitted_const = fit$const,
      fitted_lin   = predict(fit$lin),
      fitted_quad  = predict(fit$quad),
      fitted_exp   = predict(fit$exp),

      err_const = mth_vol - fitted_const,
      err_lin   = mth_vol - fitted_lin,
      err_quad  = mth_vol - fitted_quad,
      err_exp   = mth_vol - fitted_exp,

      abs_err_const = abs(err_const),
      abs_err_lin   = abs(err_lin),
      abs_err_quad  = abs(err_quad),
      abs_err_exp   = abs(err_exp)
    ) %>%
    select(-c(3:7))
}

vol_errs_tbl("2024-05-27")

```

```

## # A tibble: 21 × 10
##   `Obs. Date` date      err_const err_lin err_quad err_exp abs_err_const
##   <date>      <date>      <dbl>   <dbl>   <dbl>   <dbl>      <dbl>
## 1 2024-05-27 2024-04-26    0.0203 -0.0153 -0.00103 -0.0153    0.0203
## 2 2024-05-27 2024-04-29    0.0167 -0.0154 -0.00896 -0.0151    0.0167
## 3 2024-05-27 2024-04-30    0.0259 -0.00498 -0.000804 -0.00463    0.0259
## 4 2024-05-27 2024-05-01    0.0260 -0.00371 -0.00156 -0.00331    0.0260
## 5 2024-05-27 2024-05-02    0.0297  0.00114  0.00147  0.00158    0.0297
## 6 2024-05-27 2024-05-03    0.0373  0.00985  0.00855  0.0103    0.0373
## 7 2024-05-27 2024-05-06    0.0357  0.0118  0.00679  0.0123    0.0357
## 8 2024-05-27 2024-05-07    0.0306  0.00788  0.00208  0.00835    0.0306
## 9 2024-05-27 2024-05-08    0.0306  0.00905  0.00262  0.00950    0.0306
## 10 2024-05-27 2024-05-09    0.0316  0.0113  0.00441  0.0117    0.0316
## # i 11 more rows
## # i 3 more variables: abs_err_lin <dbl>, abs_err_quad <dbl>, abs_err_exp <dbl>

```

```
vol_errs_tbl("2024-06-03")
```

```
## # A tibble: 21 × 10
##   `Obs. Date` date      err_const  err_lin err_quad  err_exp abs_err_const
##   <date>      <date>      <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
## 1 2024-06-03  2024-05-03    0.0536  0.000181 -0.00440 -0.00221    0.0536
## 2 2024-06-03  2024-05-06    0.0520  0.00374   0.00155  0.00257    0.0520
## 3 2024-06-03  2024-05-07    0.0469  0.000405 -0.00110 -0.000429   0.0469
## 4 2024-06-03  2024-05-08    0.0469  0.00213   0.00125  0.00160    0.0469
## 5 2024-06-03  2024-05-09    0.0480  0.00489   0.00458  0.00465    0.0480
## 6 2024-06-03  2024-05-10    0.0438  0.00249   0.00270  0.00251    0.0438
## 7 2024-06-03  2024-05-13    0.0416  0.00545   0.00685  0.00605    0.0416
## 8 2024-06-03  2024-05-14    0.0315 -0.00299  -0.00131 -0.00225    0.0315
## 9 2024-06-03  2024-05-15    0.0271 -0.00563  -0.00372 -0.00478    0.0271
## 10 2024-06-03  2024-05-16    0.0271 -0.00390  -0.00182 -0.00296    0.0271
## # i 11 more rows
## # i 3 more variables: abs_err_lin <dbl>, abs_err_quad <dbl>, abs_err_exp <dbl>
```

```
vol_errs_tbl("2024-06-10")
```

```
## # A tibble: 21 × 10
##   `Obs. Date` date      err_const  err_lin err_quad  err_exp abs_err_const
##   <date>      <date>      <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
## 1 2024-06-10  2024-05-10    0.0525  0.00471   0.00363  0.00268    0.0525
## 2 2024-06-10  2024-05-13    0.0503  0.00738   0.00688  0.00649    0.0503
## 3 2024-06-10  2024-05-14    0.0401 -0.00116  -0.00150 -0.00173    0.0401
## 4 2024-06-10  2024-05-15    0.0358 -0.00389  -0.00408 -0.00418    0.0358
## 5 2024-06-10  2024-05-16    0.0358 -0.00225  -0.00231 -0.00228    0.0358
## 6 2024-06-10  2024-05-17    0.0325 -0.00395  -0.00388 -0.00374    0.0325
## 7 2024-06-10  2024-05-20    0.0314 -0.000110  0.000246  0.000609    0.0314
## 8 2024-06-10  2024-05-21    0.0240 -0.00589  -0.00547 -0.00505    0.0240
## 9 2024-06-10  2024-05-22    0.0238 -0.00442  -0.00394 -0.00349    0.0238
## 10 2024-06-10  2024-05-23    0.0237 -0.00294  -0.00242 -0.00195    0.0237
## # i 11 more rows
## # i 3 more variables: abs_err_lin <dbl>, abs_err_quad <dbl>, abs_err_exp <dbl>
```

```
vol_errs_tbl("2024-06-17")
```

```
## # A tibble: 21 × 10
##   `Obs. Date` date      err_const  err_lin err_quad  err_exp abs_err_const
##   <date>      <date>      <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
## 1 2024-06-17  2024-05-17    0.0353  0.000264 -0.00200 -0.00112    0.0353
## 2 2024-06-17  2024-05-20    0.0342  0.00294   0.00189  0.00230    0.0342
## 3 2024-06-17  2024-05-21    0.0268 -0.00323  -0.00393 -0.00366    0.0268
## 4 2024-06-17  2024-05-22    0.0267 -0.00214  -0.00253 -0.00239    0.0267
## 5 2024-06-17  2024-05-23    0.0265 -0.00105  -0.00115 -0.00112    0.0265
## 6 2024-06-17  2024-05-24    0.0278  0.00149   0.00165  0.00157    0.0278
## 7 2024-06-17  2024-05-28    0.0255  0.00409   0.00500  0.00460    0.0255
## 8 2024-06-17  2024-05-29    0.0264  0.00630   0.00731  0.00687    0.0264
## 9 2024-06-17  2024-05-30    0.0292  0.0103   0.0114  0.0109    0.0292
## 10 2024-06-17  2024-05-31    0.0135 -0.00413  -0.00298 -0.00348    0.0135
## # i 11 more rows
## # i 3 more variables: abs_err_lin <dbl>, abs_err_quad <dbl>, abs_err_exp <dbl>
```

```
vol_errs_tbl("2024-06-24")
```

```
## # A tibble: 21 × 10
##   `Obs. Date` date      err_const err_lin err_quad err_exp abs_err_const
##   <date>      <date>      <dbl>    <dbl>    <dbl>    <dbl>      <dbl>
## 1 2024-06-24 2024-05-23  0.0243  0.00130 -0.00568 -0.000187  0.0243
## 2 2024-06-24 2024-05-24  0.0256  0.00358 -0.00212  0.00231   0.0256
## 3 2024-06-24 2024-05-28  0.0233  0.00514  0.00374  0.00462   0.0233
## 4 2024-06-24 2024-05-29  0.0242  0.00708  0.00654  0.00672   0.0242
## 5 2024-06-24 2024-05-30  0.0270  0.0108  0.0110  0.0106   0.0270
## 6 2024-06-24 2024-05-31  0.0113 -0.00387 -0.00293 -0.00397   0.0113
## 7 2024-06-24 2024-06-03  0.00931 -0.00295 -0.000425 -0.00273   0.00931
## 8 2024-06-24 2024-06-04  0.00613 -0.00514 -0.00226 -0.00485   0.00613
## 9 2024-06-24 2024-06-05  0.00515 -0.00514 -0.00198 -0.00477   0.00515
## 10 2024-06-24 2024-06-06  0.000228 -0.00908 -0.00573 -0.00866   0.000228
## # i 11 more rows
## # i 3 more variables: abs_err_lin <dbl>, abs_err_quad <dbl>, abs_err_exp <dbl>
```

```
vol_errs_tbl("2024-07-01")
```

```
## # A tibble: 21 × 10
##   `Obs. Date` date      err_const err_lin err_quad err_exp abs_err_const
##   <date>      <date>      <dbl>    <dbl>    <dbl>    <dbl>      <dbl>
## 1 2024-07-01 2024-05-31  0.0246  0.00307  0.00347  2.79e-3   0.0246
## 2 2024-07-01 2024-06-03  0.0226  0.00269  0.00288  2.57e-3   0.0226
## 3 2024-07-01 2024-06-04  0.0194  0.0000588  0.000189 -1.69e-5   0.0194
## 4 2024-07-01 2024-06-05  0.0184 -0.000370 -0.000296 -4.06e-4   0.0184
## 5 2024-07-01 2024-06-06  0.0135 -0.00475 -0.00473 -4.75e-3   0.0135
## 6 2024-07-01 2024-06-07  0.0140 -0.00373 -0.00375 -3.70e-3   0.0140
## 7 2024-07-01 2024-06-10  0.0139 -0.00215 -0.00228 -2.04e-3   0.0139
## 8 2024-07-01 2024-06-11  0.0131 -0.00245 -0.00261 -2.33e-3   0.0131
## 9 2024-07-01 2024-06-12  0.0165  0.00155  0.00137  1.68e-3   0.0165
## 10 2024-07-01 2024-06-13  0.0163  0.00184  0.00165  1.98e-3   0.0163
## # i 11 more rows
## # i 3 more variables: abs_err_lin <dbl>, abs_err_quad <dbl>, abs_err_exp <dbl>
```

```
vol_errs_tbl("2024-07-08")
```



```
## # A tibble: 21 × 10
##   `Obs. Date` date      err_const  err_lin  err_quad  err_exp abs_err_const
##   <date>      <date>      <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
## 1 2024-07-08 2024-06-06    0.0192 -0.00555 -0.000291 -5.66e-3  0.0192
## 2 2024-07-08 2024-06-07    0.0197 -0.00436 -0.000139 -4.40e-3  0.0197
## 3 2024-07-08 2024-06-10    0.0196 -0.00226 -0.000739 -2.12e-3  0.0196
## 4 2024-07-08 2024-06-11    0.0188 -0.00239 -0.00163 -2.20e-3  0.0188
## 5 2024-07-08 2024-06-12    0.0222  0.00178  0.00185  2.01e-3  0.0222
## 6 2024-07-08 2024-06-13    0.0220  0.00224  0.00169  2.50e-3  0.0220
## 7 2024-07-08 2024-06-14    0.0216  0.00258  0.00147  2.86e-3  0.0216
## 8 2024-07-08 2024-06-17    0.0168 -0.0000552 -0.00241  2.63e-4  0.0168
## 9 2024-07-08 2024-06-18    0.0159 -0.000229 -0.00286  8.86e-5  0.0159
## 10 2024-07-08 2024-06-20    0.0171  0.00244 -0.000535  2.73e-3  0.0171
## # i 11 more rows
## # i 3 more variables: abs_err_lin <dbl>, abs_err_quad <dbl>, abs_err_exp <dbl>
```

```
vol_errs_tbl("2024-07-15")
```

```
## # A tibble: 21 × 10
##   `Obs. Date` date      err_const  err_lin  err_quad  err_exp abs_err_const
##   <date>      <date>      <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
## 1 2024-07-15 2024-06-13    0.0140  0.000681 -0.00280 -0.000159  0.0140
## 2 2024-07-15 2024-06-14    0.0136  0.00103 -0.00179  0.000331  0.0136
## 3 2024-07-15 2024-06-17    0.00880 -0.00154 -0.00263 -0.00188  0.00880
## 4 2024-07-15 2024-06-18    0.00791 -0.00170 -0.00230 -0.00193  0.00791
## 5 2024-07-15 2024-06-20    0.00914  0.00100  0.00125  0.000953  0.00914
## 6 2024-07-15 2024-06-21    0.00977  0.00237  0.00298  0.00240  0.00977
## 7 2024-07-15 2024-06-24    0.0110  0.00580  0.00723  0.00601  0.0110
## 8 2024-07-15 2024-06-25    0.0104  0.00594  0.00756  0.00620  0.0104
## 9 2024-07-15 2024-06-26    0.00420  0.000487  0.00225  0.000778  0.00420
## 10 2024-07-15 2024-06-27    0.00193 -0.00104  0.000817 -0.000724  0.00193
## # i 11 more rows
## # i 3 more variables: abs_err_lin <dbl>, abs_err_quad <dbl>, abs_err_exp <dbl>
```

```
vol_errs_tbl("2024-07-22")
```

```
## # A tibble: 21 × 10
##   `Obs. Date` date      err_const  err_lin  err_quad  err_exp abs_err_const
##   <date>      <date>      <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
## 1 2024-07-22 2024-06-21   -0.0193  0.0121 -0.00986  0.0127  0.0193
## 2 2024-07-22 2024-06-24   -0.0180  0.0123  0.00211  0.0128  0.0180
## 3 2024-07-22 2024-06-25   -0.0186  0.0113  0.00453  0.0119  0.0186
## 4 2024-07-22 2024-06-26   -0.0248  0.00479  0.00108  0.00528  0.0248
## 5 2024-07-22 2024-06-27   -0.0271  0.00217  0.00127  0.00263  0.0271
## 6 2024-07-22 2024-06-28   -0.0247  0.00415  0.00578  0.00459  0.0247
## 7 2024-07-22 2024-07-01   -0.0313 -0.00346  0.00406 -0.00312  0.0313
## 8 2024-07-22 2024-07-02   -0.0356 -0.00809  0.000819 -0.00779  0.0356
## 9 2024-07-22 2024-07-03   -0.0380 -0.0108 -0.000804 -0.0106  0.0380
## 10 2024-07-22 2024-07-05   -0.0371 -0.0107  0.000695 -0.0105  0.0371
## # i 11 more rows
## # i 3 more variables: abs_err_lin <dbl>, abs_err_quad <dbl>, abs_err_exp <dbl>
```

```
vol_errs_tbl("2024-07-29")
```

```
## # A tibble: 21 × 10
##   `Obs. Date` date      err_const  err_lin  err_quad  err_exp abs_err_const
##   <date>      <date>      <dbl>    <dbl>    <dbl>    <dbl>      <dbl>
## 1 2024-07-29 2024-06-28 -0.0592  0.0294   0.00279  0.0272     0.0592
## 2 2024-07-29 2024-07-01 -0.0657  0.0154   0.00260  0.0157     0.0657
## 3 2024-07-29 2024-07-02 -0.0700  0.00866 -0.000214 0.00959     0.0700
## 4 2024-07-29 2024-07-03 -0.0724  0.00380 -0.00146  0.00536     0.0724
## 5 2024-07-29 2024-07-05 -0.0716 -0.000322 0.000657 0.00230     0.0716
## 6 2024-07-29 2024-07-08 -0.0715 -0.00764 0.000241 -0.00392     0.0715
## 7 2024-07-29 2024-07-09 -0.0809 -0.0196  -0.0101  -0.0156     0.0809
## 8 2024-07-29 2024-07-10 -0.0746 -0.0157  -0.00486 -0.0116     0.0746
## 9 2024-07-29 2024-07-11 -0.0645 -0.00806 0.00378  -0.00391     0.0645
## 10 2024-07-29 2024-07-12 -0.0635 -0.00954 0.00297  -0.00540     0.0635
## # i 11 more rows
## # i 3 more variables: abs_err_lin <dbl>, abs_err_quad <dbl>, abs_err_exp <dbl>
```

Summarize volatility model performance across all dates

```

obs_dates <- c(
  "2024-05-27", "2024-06-03", "2024-06-10", "2024-06-17",
  "2024-06-24", "2024-07-01", "2024-07-08", "2024-07-15",
  "2024-07-22", "2024-07-29"
)

vol_summary_tbl <- map_dfr(obs_dates, function(d) {
  vol_errs_tbl(d) %>%
    summarise(
      Date      = d,
      RMSE_const = sqrt(mean(err_const^2, na.rm = TRUE)),
      RMSE_lin   = sqrt(mean(err_lin^2,   na.rm = TRUE)),
      RMSE_quad  = sqrt(mean(err_quad^2,  na.rm = TRUE)),
      RMSE_exp   = sqrt(mean(err_exp^2,   na.rm = TRUE)),
      MAE_const  = mean(abs_err_const, na.rm = TRUE),
      MAE_lin    = mean(abs_err_lin,   na.rm = TRUE),
      MAE_quad   = mean(abs_err_quad,  na.rm = TRUE),
      MAE_exp    = mean(abs_err_exp,   na.rm = TRUE)
    )
}) %>%
  bind_rows(
    summarise(.,
      Date      = "Mean",
      RMSE_const = round(mean(RMSE_const), 4),
      RMSE_lin   = mean(RMSE_lin),
      RMSE_quad  = mean(RMSE_quad),
      RMSE_exp   = mean(RMSE_exp),
      MAE_const  = mean(MAE_const),
      MAE_lin    = mean(MAE_lin),
      MAE_quad   = mean(MAE_quad),
      MAE_exp    = mean(MAE_exp)
    )
  )

vol_summary_tbl

```

```

## # A tibble: 11 × 9
##   Date      RMSE_const RMSE_lin RMSE_quad RMSE_exp MAE_const MAE_lin MAE_quad
##   <chr>      <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
## 1 2024-05-27  0.0223  0.00795  0.00487  0.00822  0.0186  0.00662  0.00398
## 2 2024-06-03  0.0317  0.00459  0.00408  0.00424  0.0274  0.00381  0.00346
## 3 2024-06-10  0.0275  0.00439  0.00436  0.00436  0.0227  0.00349  0.00345
## 4 2024-06-17  0.0196  0.00441  0.00428  0.00431  0.0155  0.00378  0.00352
## 5 2024-06-24  0.0128  0.00519  0.00388  0.00490  0.00835  0.00425  0.00294
## 6 2024-07-01  0.0143  0.00273  0.00273  0.00274  0.0131  0.00229  0.00229
## 7 2024-07-08  0.0158  0.00376  0.00252  0.00395  0.0139  0.00311  0.00201
## 8 2024-07-15  0.00888  0.00528  0.00497  0.00519  0.00755  0.00420  0.00419
## 9 2024-07-22  0.0284  0.0114  0.00522  0.0114  0.0258  0.0102  0.00417
## 10 2024-07-29  0.0556  0.0139  0.00689  0.0115  0.0491  0.0118  0.00559
## 11 Mean      0.0237  0.00637  0.00438  0.00608  0.0202  0.00536  0.00356
## # i 1 more variable: MAE_exp <dbl>

```

Best-fit model per date for volatility

```

vol_best_fit_tbl <- vol_summary_tbl %>%
  rowwise() %>%
  mutate(
    RMSE_Best_Fit = c("Constant", "Linear", "Quadratic", "Exponential")[
      which.min(c(RMSE_const, RMSE_lin, RMSE_quad, RMSE_exp))
    ],
    RMSE = min(RMSE_const, RMSE_lin, RMSE_quad, RMSE_exp) %>% round(4),

    MAE_Best_Fit = c("Constant", "Linear", "Quadratic", "Exponential")[
      which.min(c(MAE_const, MAE_lin, MAE_quad, MAE_exp))
    ],
    MAE = min(MAE_const, MAE_lin, MAE_quad, MAE_exp) %>% round(4)
  ) %>%
  ungroup() %>%
  select(Date, RMSE_Best_Fit, RMSE, MAE_Best_Fit, MAE)

vol_best_fit_tbl

```

```

## # A tibble: 11 × 5
##   Date      RMSE_Best_Fit  RMSE MAE_Best_Fit  MAE
##   <chr>      <chr>          <dbl> <chr>          <dbl>
## 1 2024-05-27 Quadratic    0.0049 Quadratic    0.004
## 2 2024-06-03 Quadratic    0.0041 Quadratic    0.0035
## 3 2024-06-10 Quadratic    0.0044 Exponential  0.0034
## 4 2024-06-17 Quadratic    0.0043 Quadratic    0.0035
## 5 2024-06-24 Quadratic    0.0039 Quadratic    0.0029
## 6 2024-07-01 Quadratic    0.0027 Exponential  0.0023
## 7 2024-07-08 Quadratic    0.0025 Quadratic    0.002
## 8 2024-07-15 Quadratic    0.005  Exponential  0.0041
## 9 2024-07-22 Quadratic    0.0052 Quadratic    0.0042
## 10 2024-07-29 Quadratic    0.0069 Quadratic    0.0056
## 11 Mean      Quadratic    0.0044 Quadratic    0.0036

```

Function to create a plot for call prices on a given date

```

plot_call_prices <- function(obs_date) {
  # Build volatility + call price tables
  vol_dat  <- hist_vol(obs_date) %>% fit_vol()
  calls_dat <- call_tbl_fun(obs_date, vol_dat)

  # Build x-axis breaks
  x_min    <- min(calls_dat$x_day)
  x_max    <- max(calls_dat$x_day)
  x_breaks <- seq(x_min, x_max, length.out = 5)
  x_labels <- format(as.Date(obs_date) + round(x_breaks), "%b %d")

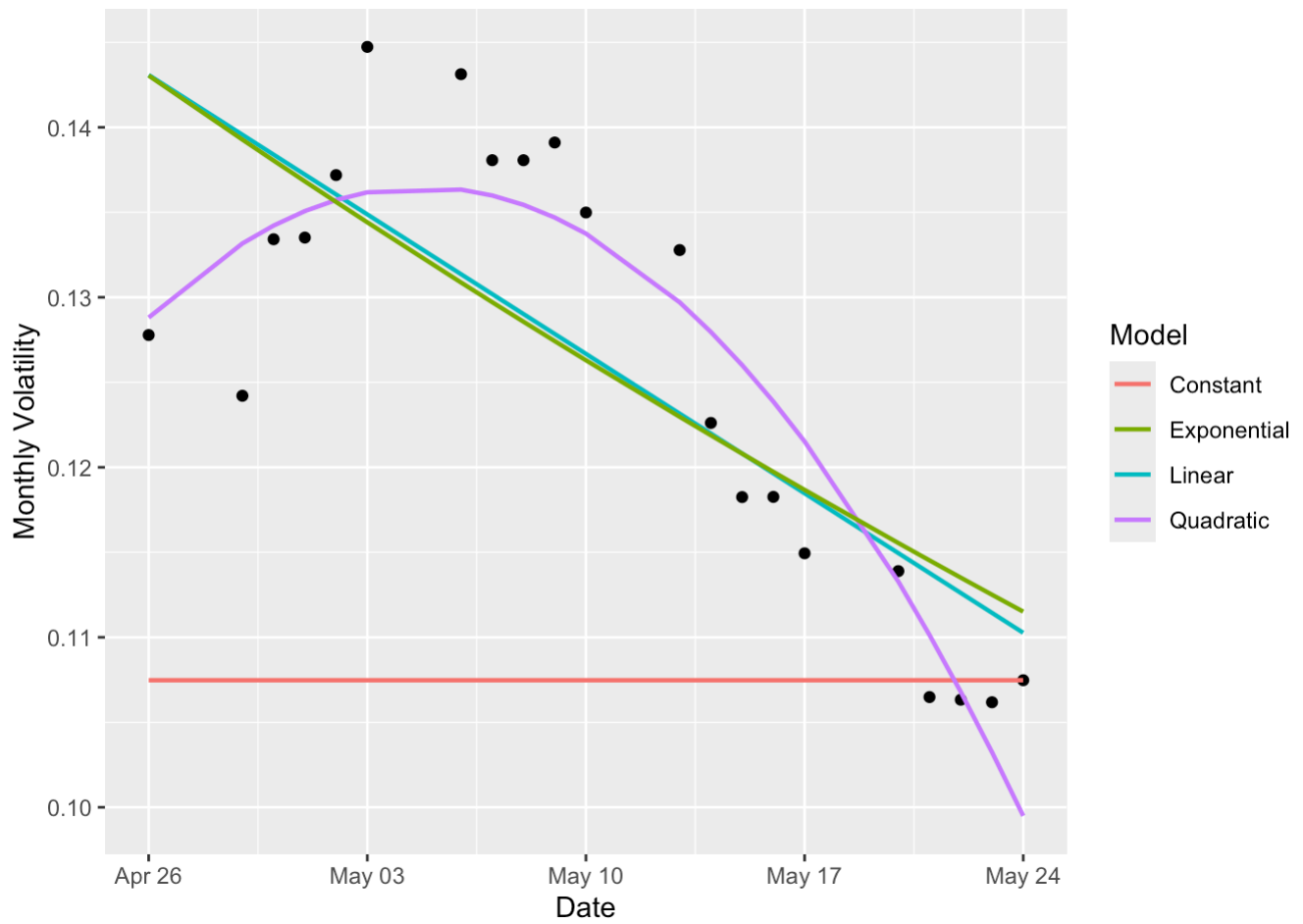
  # Get ending date for title
  end_date <- max(as.Date(calls_dat$`Exp. Date`))

  ggplot(calls_dat, aes(x = x_day, y = `True Call`)) +
    geom_point(size = 1.5) +
    geom_line(aes(y = `Const Call`, color = "Constant"), linewidth = 0.8) +
    geom_line(aes(y = `Lin Call`, color = "Linear"), linewidth = 0.8) +
    geom_line(aes(y = `Quad Call`, color = "Quadratic"), linewidth = 0.8) +
    geom_line(aes(y = `Exp Call`, color = "Exponential"), linewidth = 0.8) +
    scale_x_continuous(
      breaks = x_breaks,
      labels = x_labels
    ) +
    labs(
      # title = paste(
      #   "Call Prices from",
      #   format(as.Date(obs_date), "%B %d"),
      #   "to",
      #   format(end_date, "%B %d, %Y")
      # ),
      x = "Date",
      y = "Call Price ($)",
      color = "Model"
    ) +
    theme(plot.title = element_text(hjust = 0.5))
}

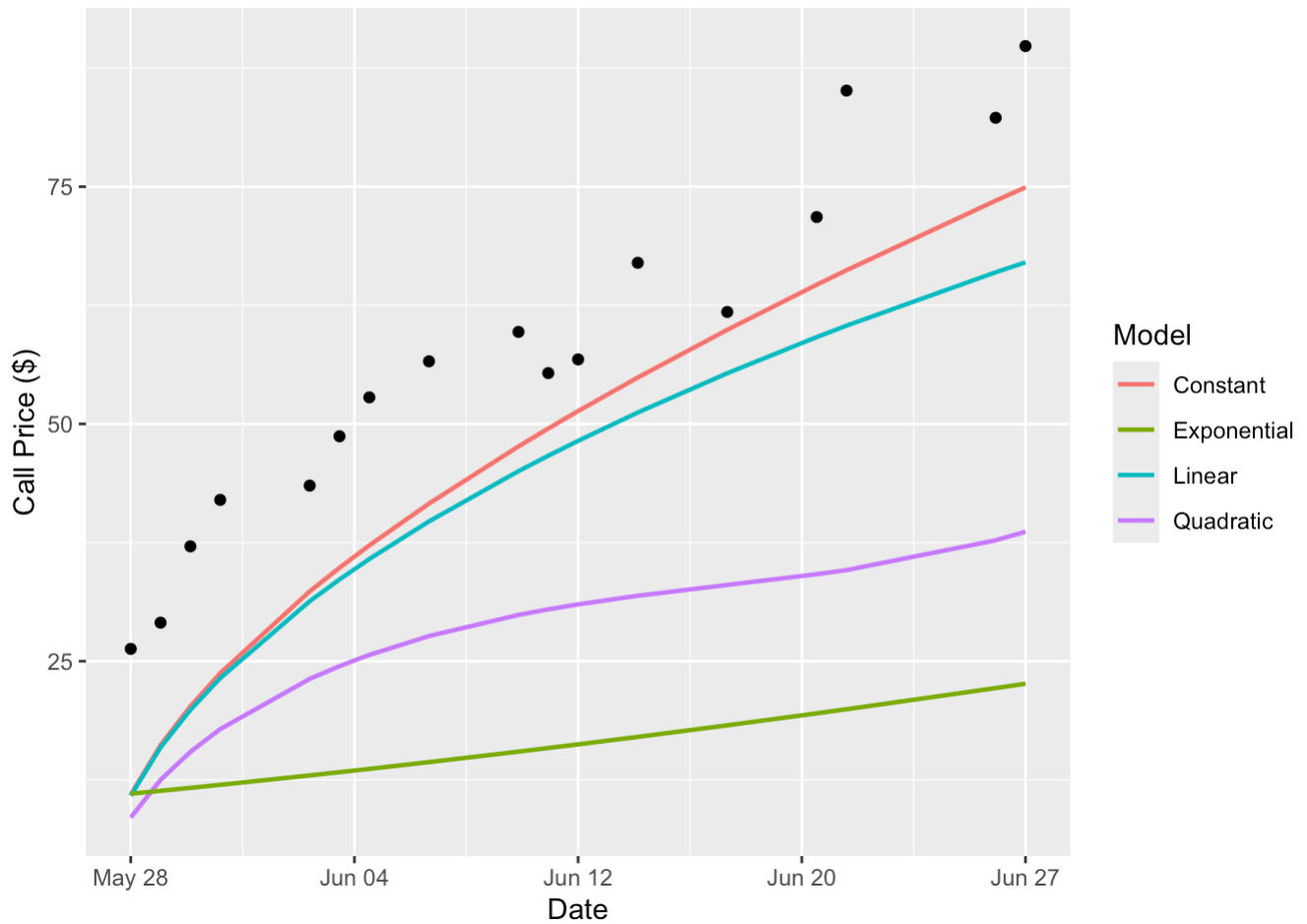
```

Plot all volatilities and call prices

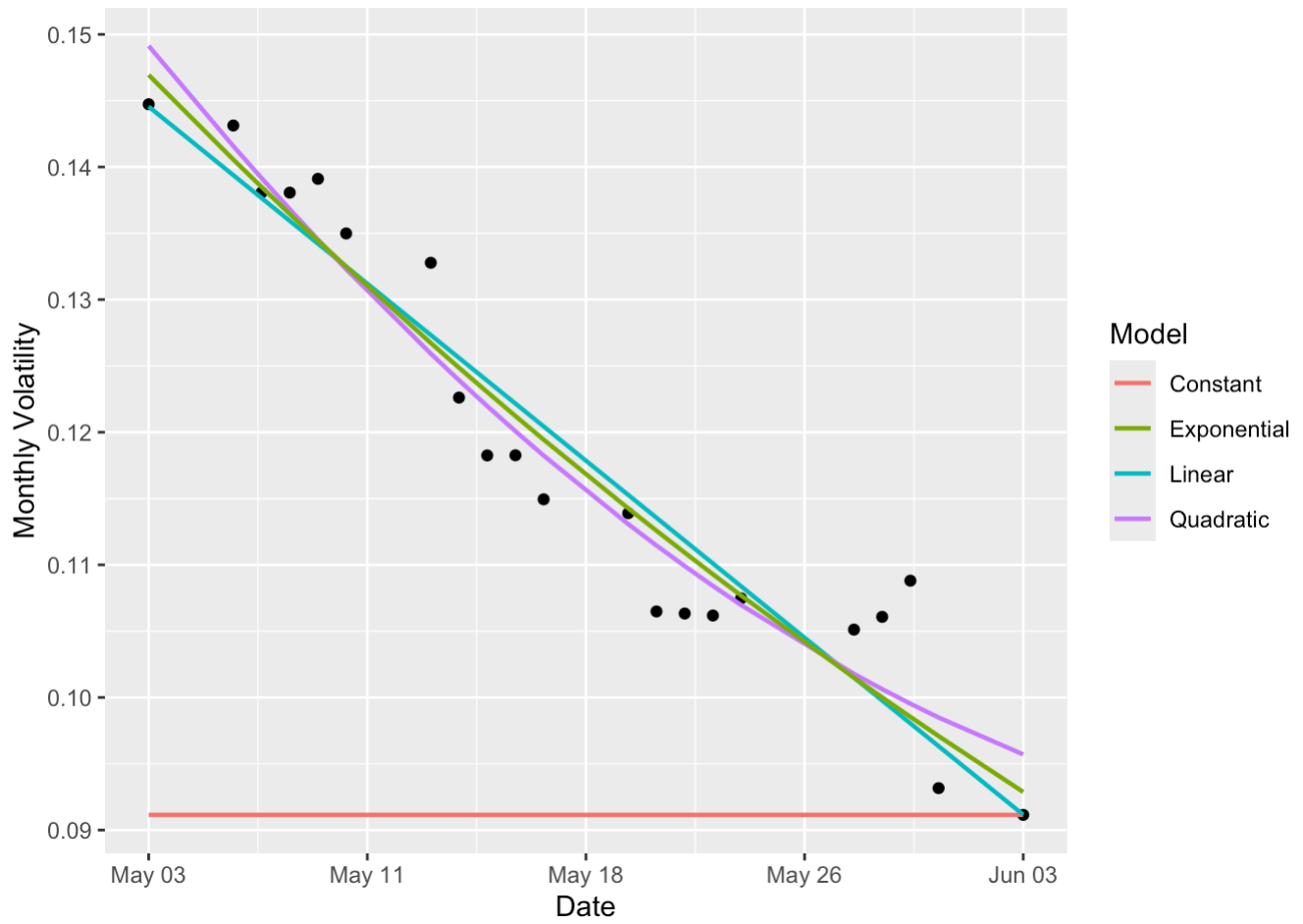
```
plot_vol("2024-05-27")
```



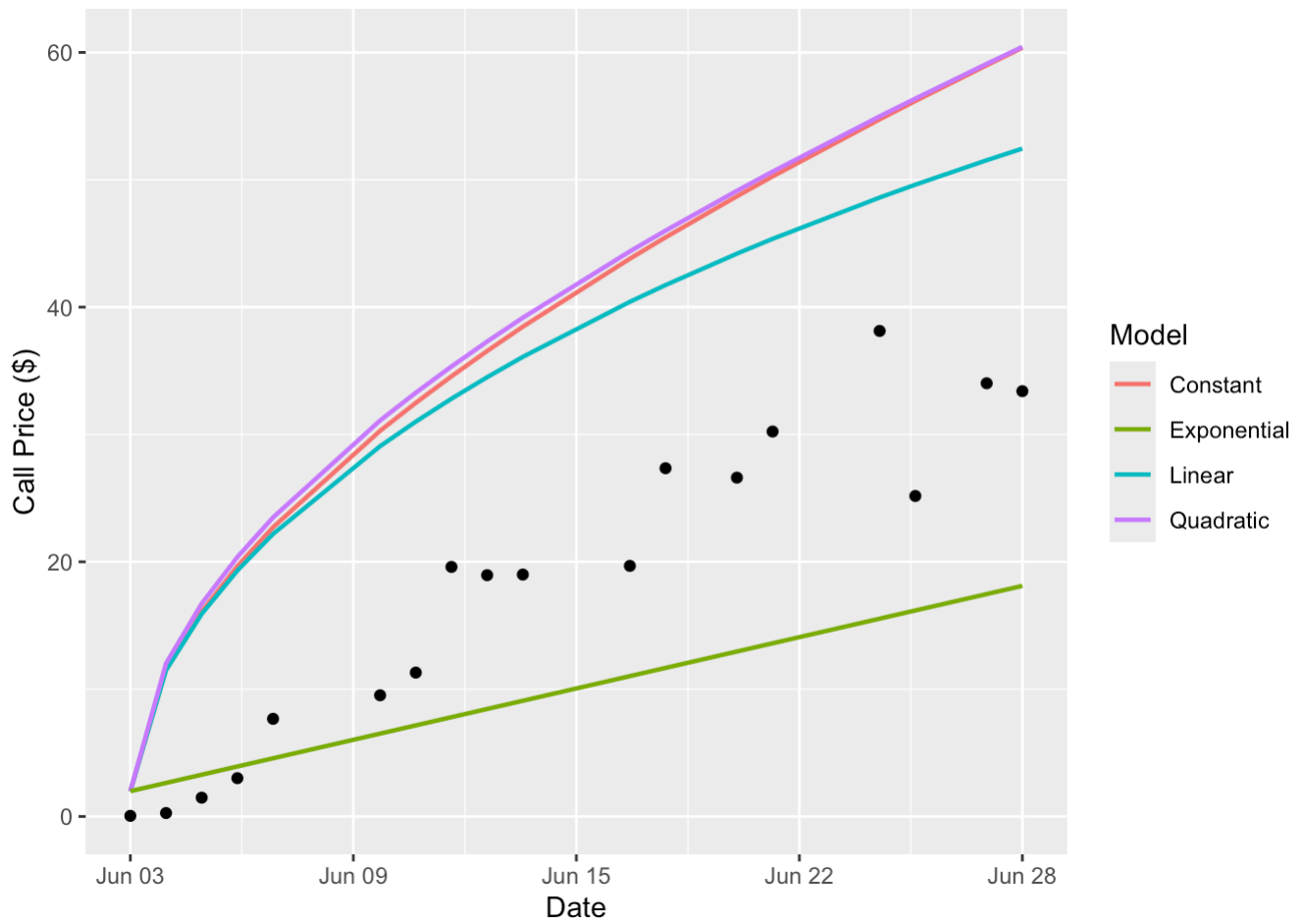
```
plot_call_prices("2024-05-27")
```



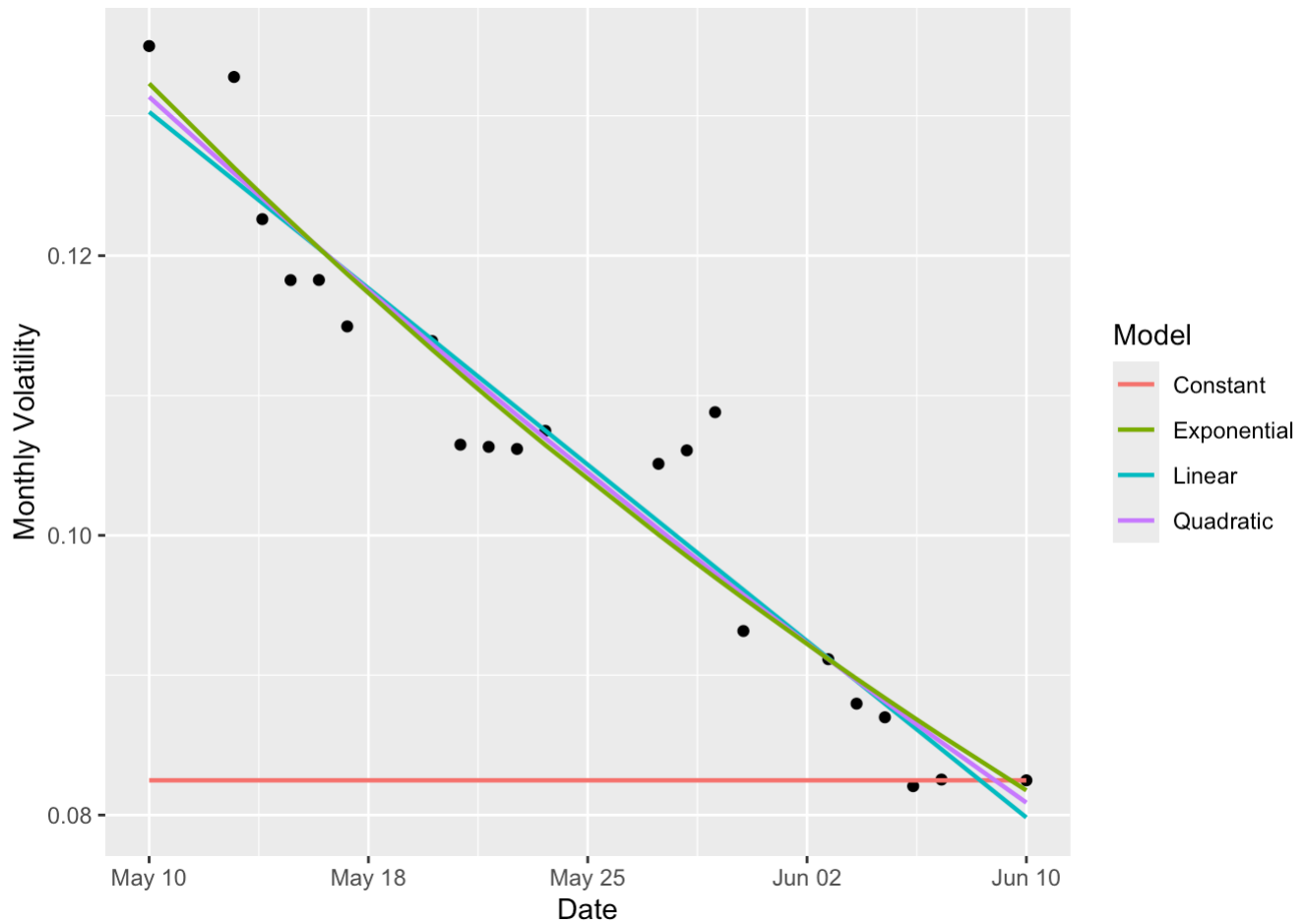
```
plot_vol("2024-06-03")
```



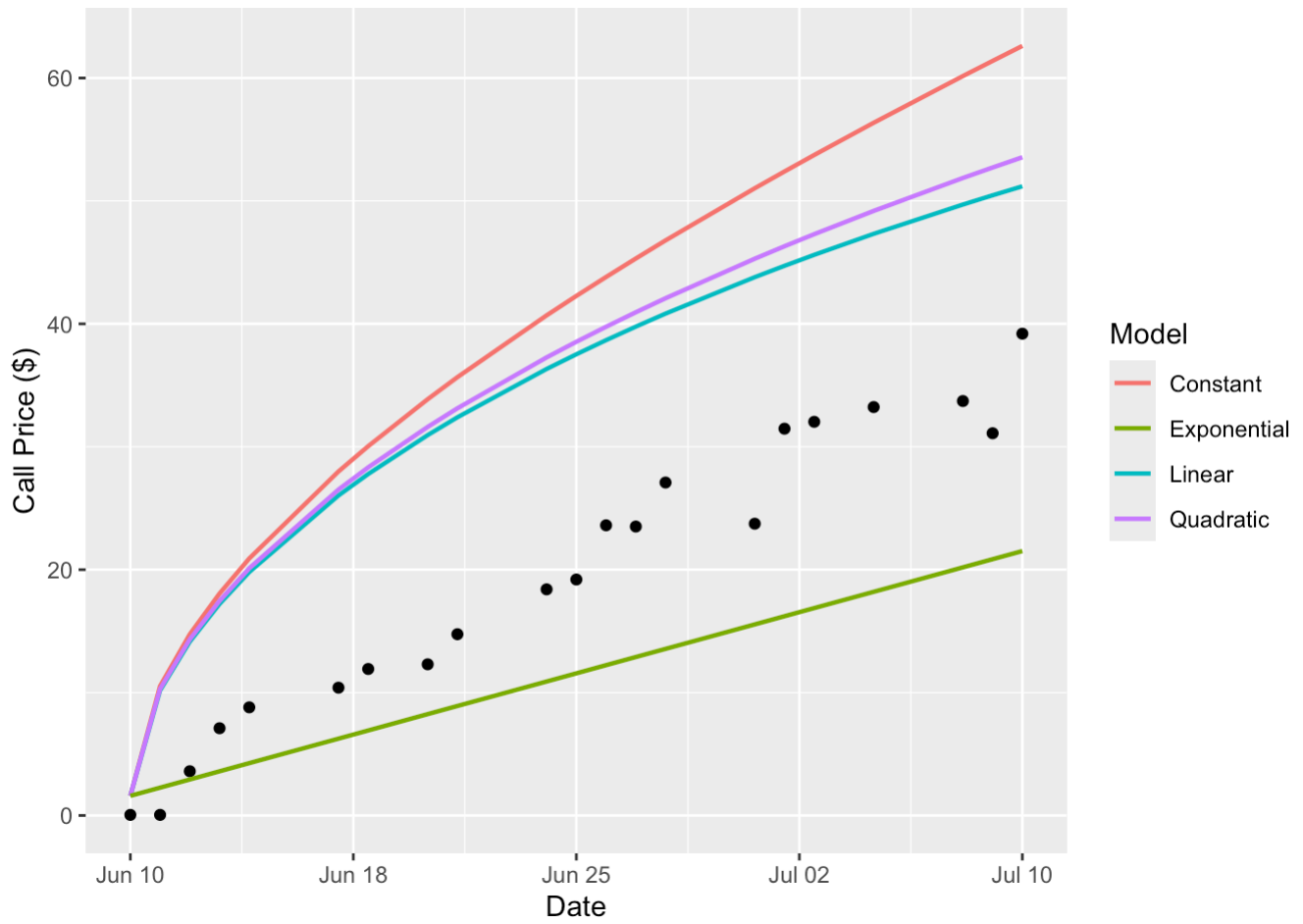
```
plot_call_prices("2024-06-03")
```



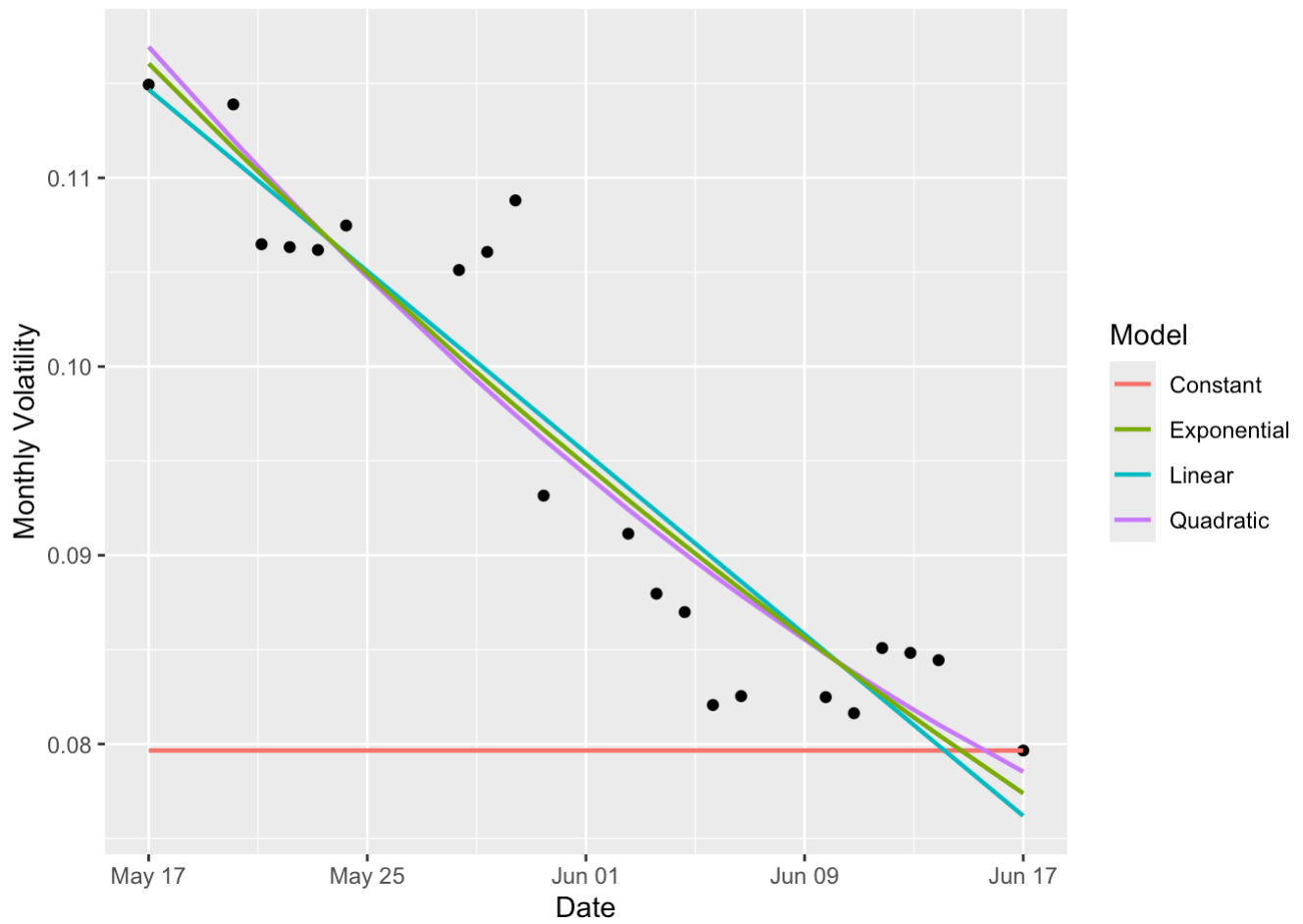
plot_vol("2024-06-10")



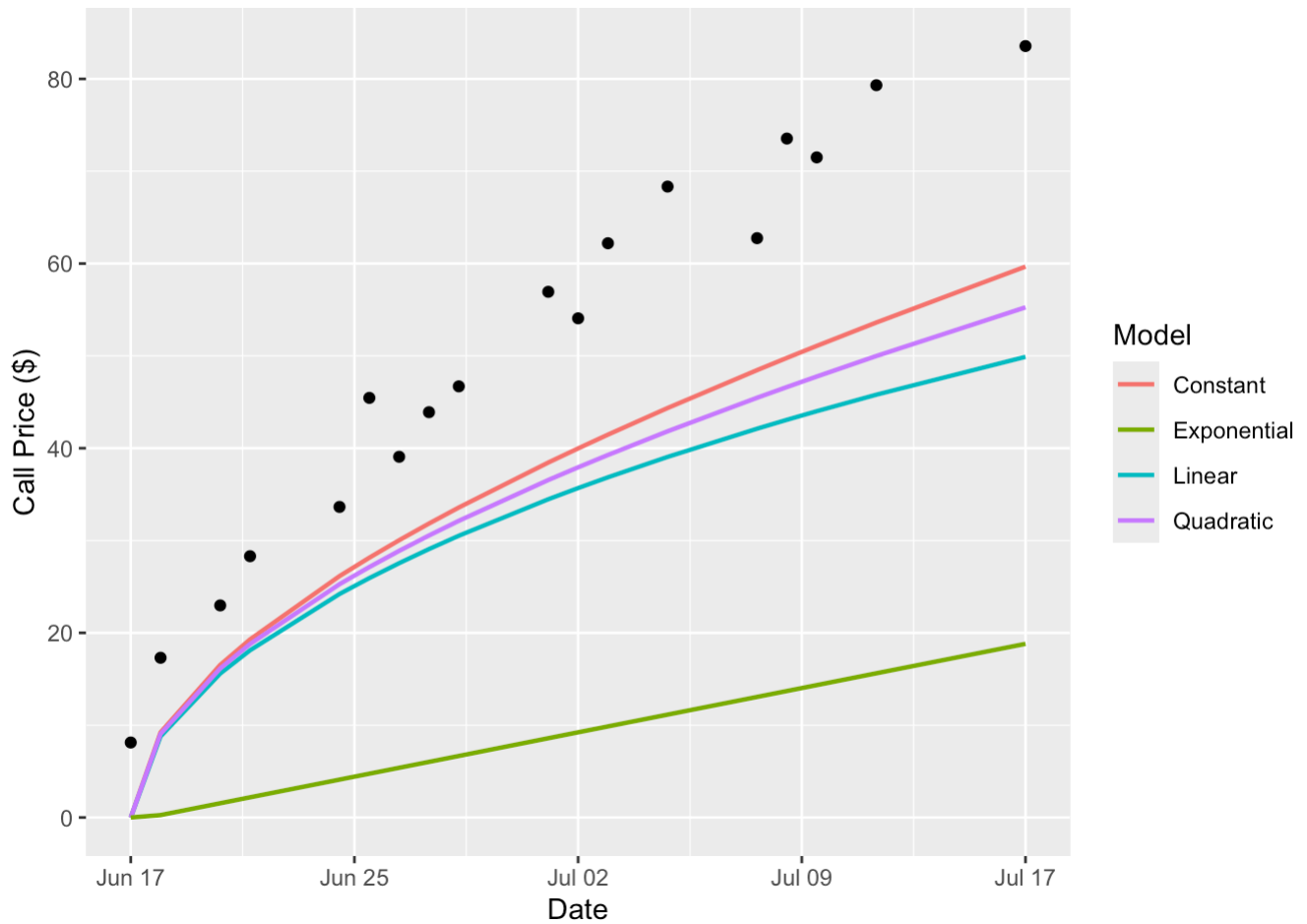

```
plot_call_prices("2024-06-10")
```



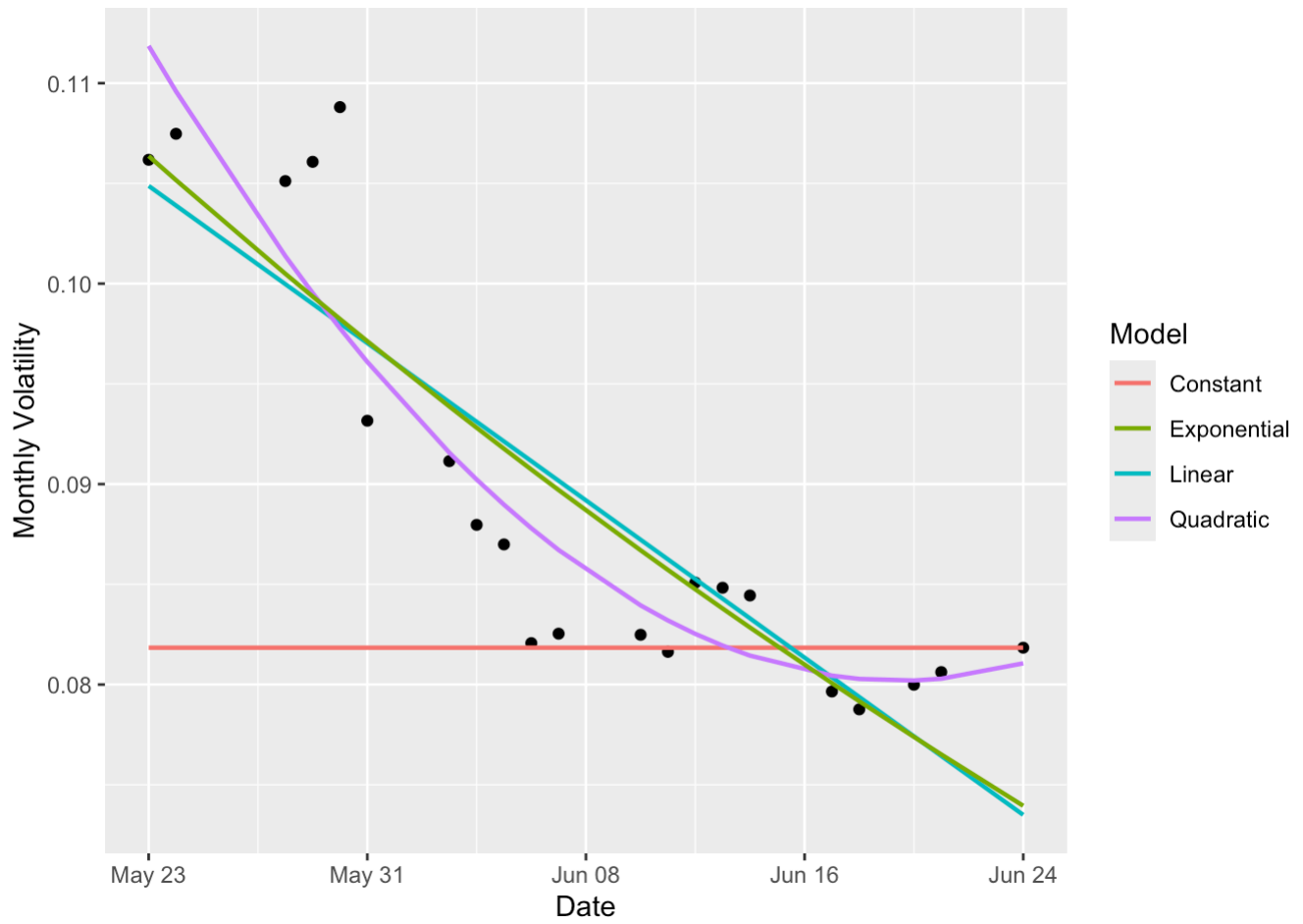
```
plot_vol("2024-06-17")
```



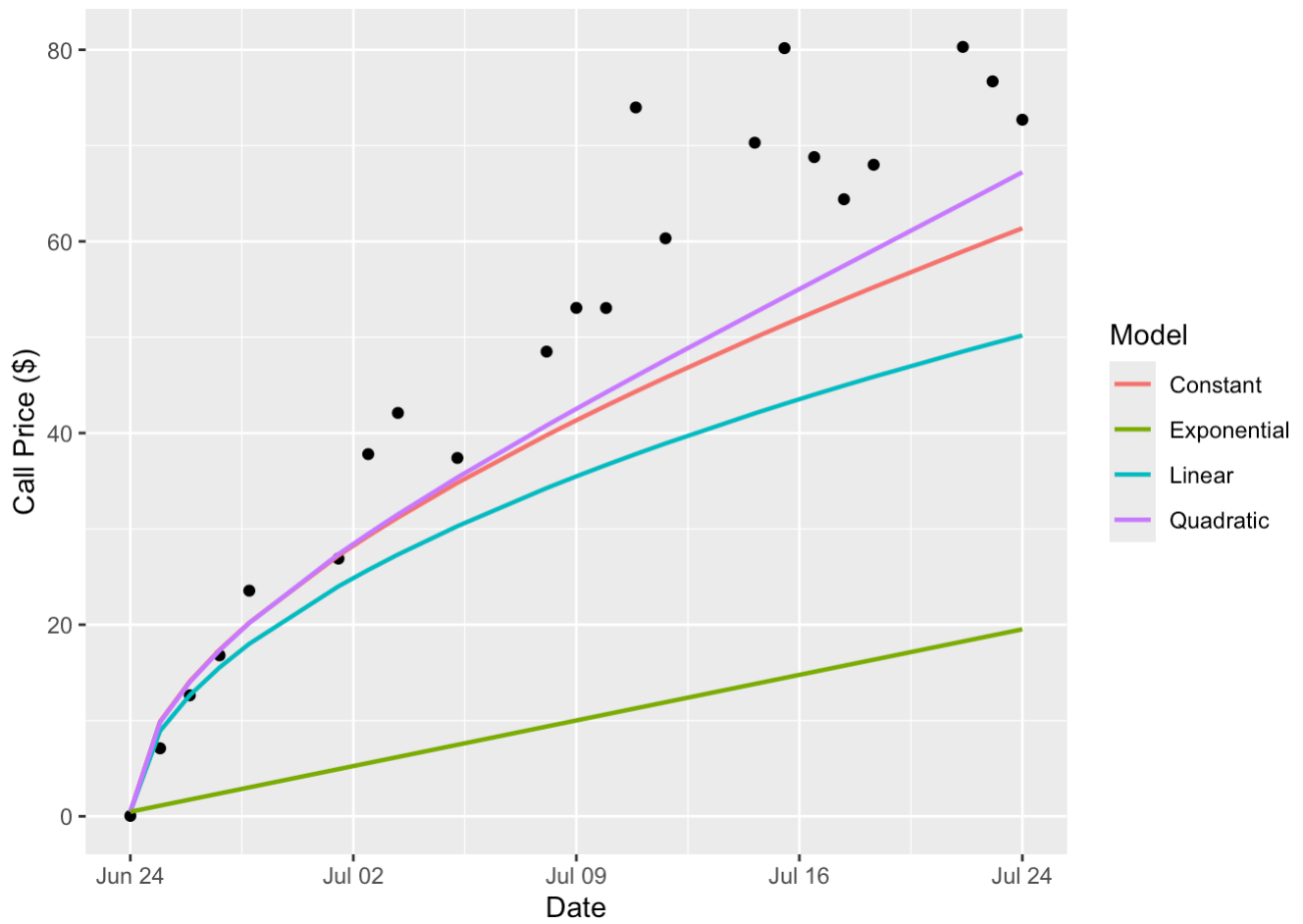
```
plot_call_prices("2024-06-17")
```



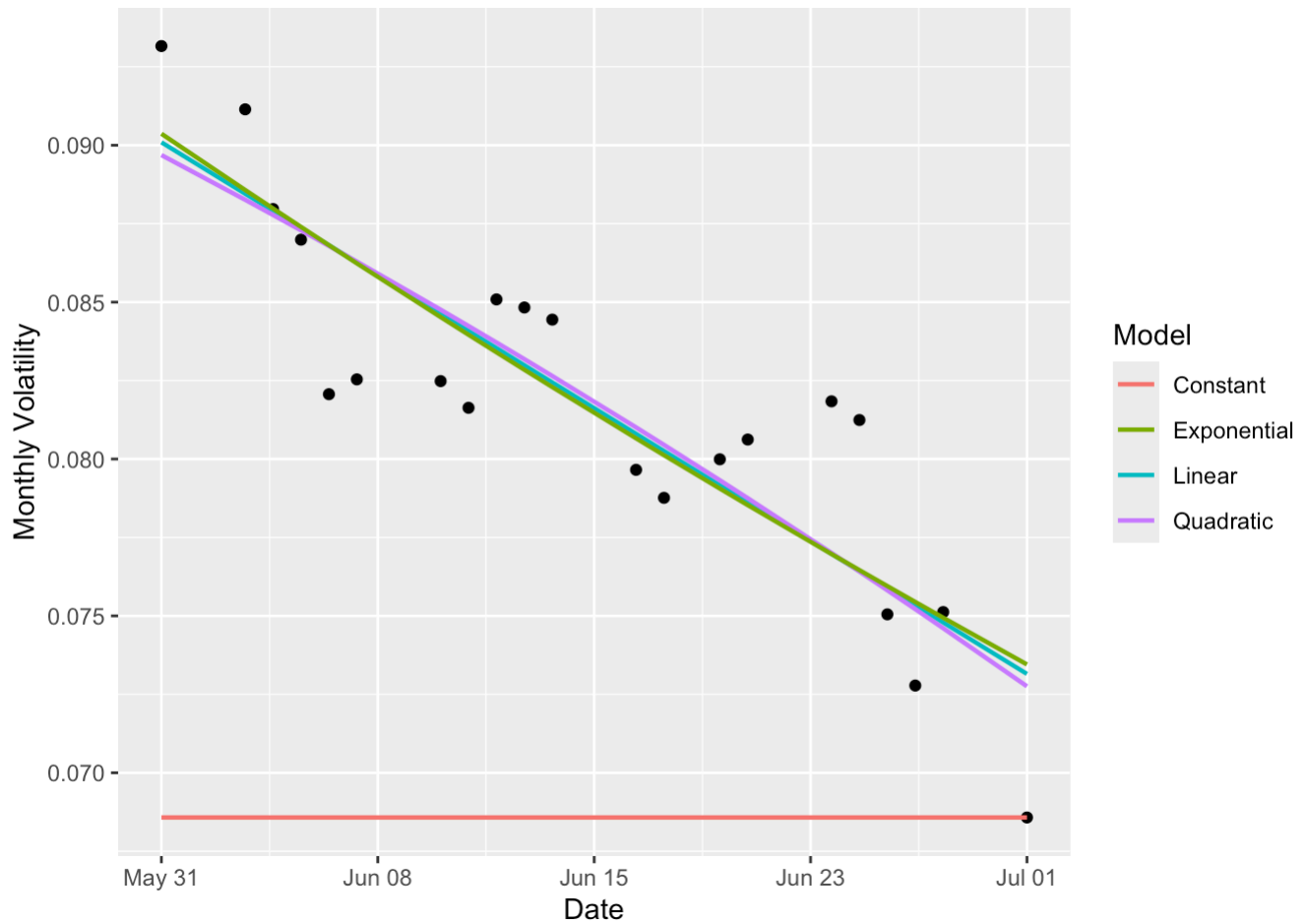
```
plot_vol("2024-06-24")
```



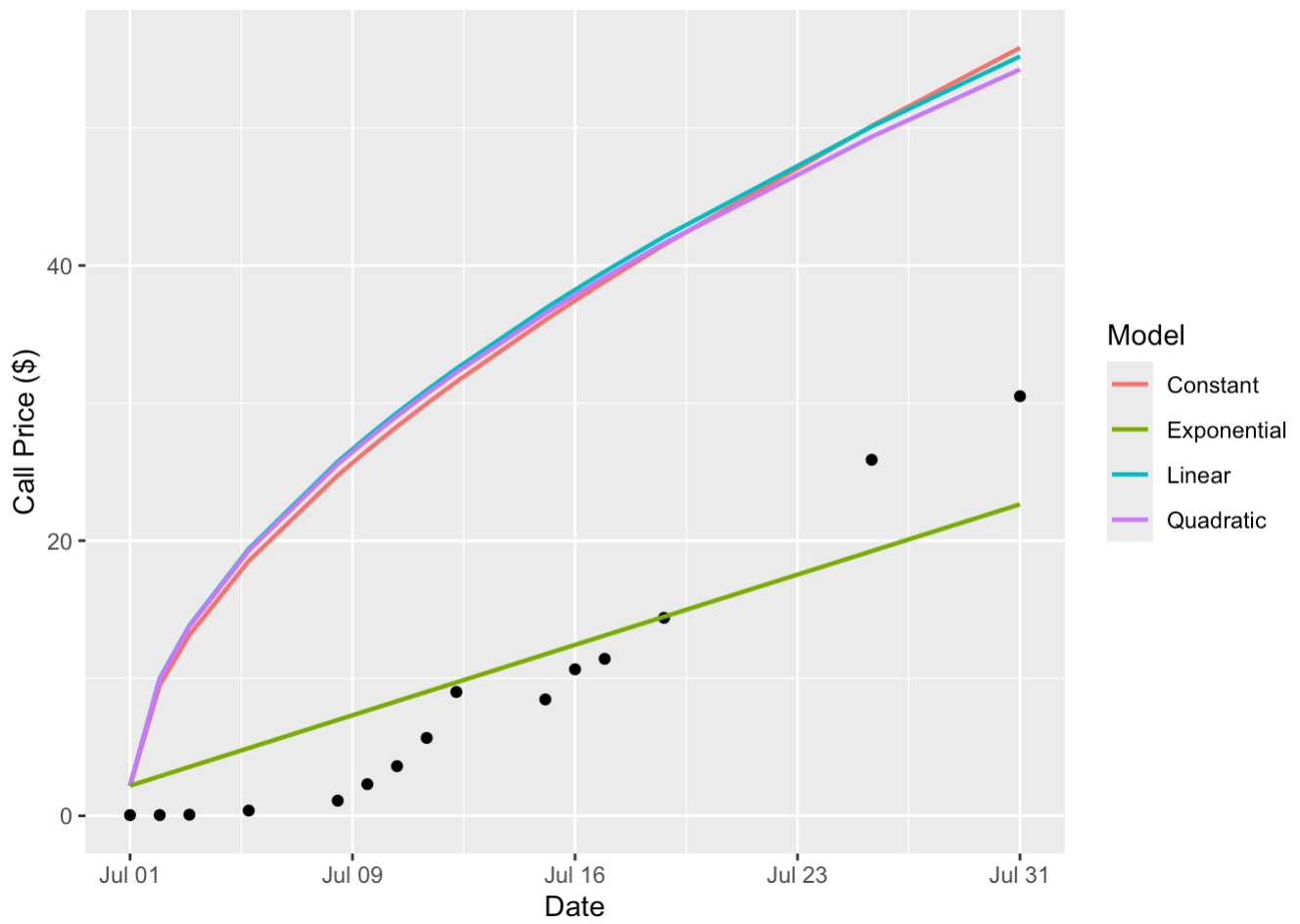
```
plot_call_prices("2024-06-24")
```



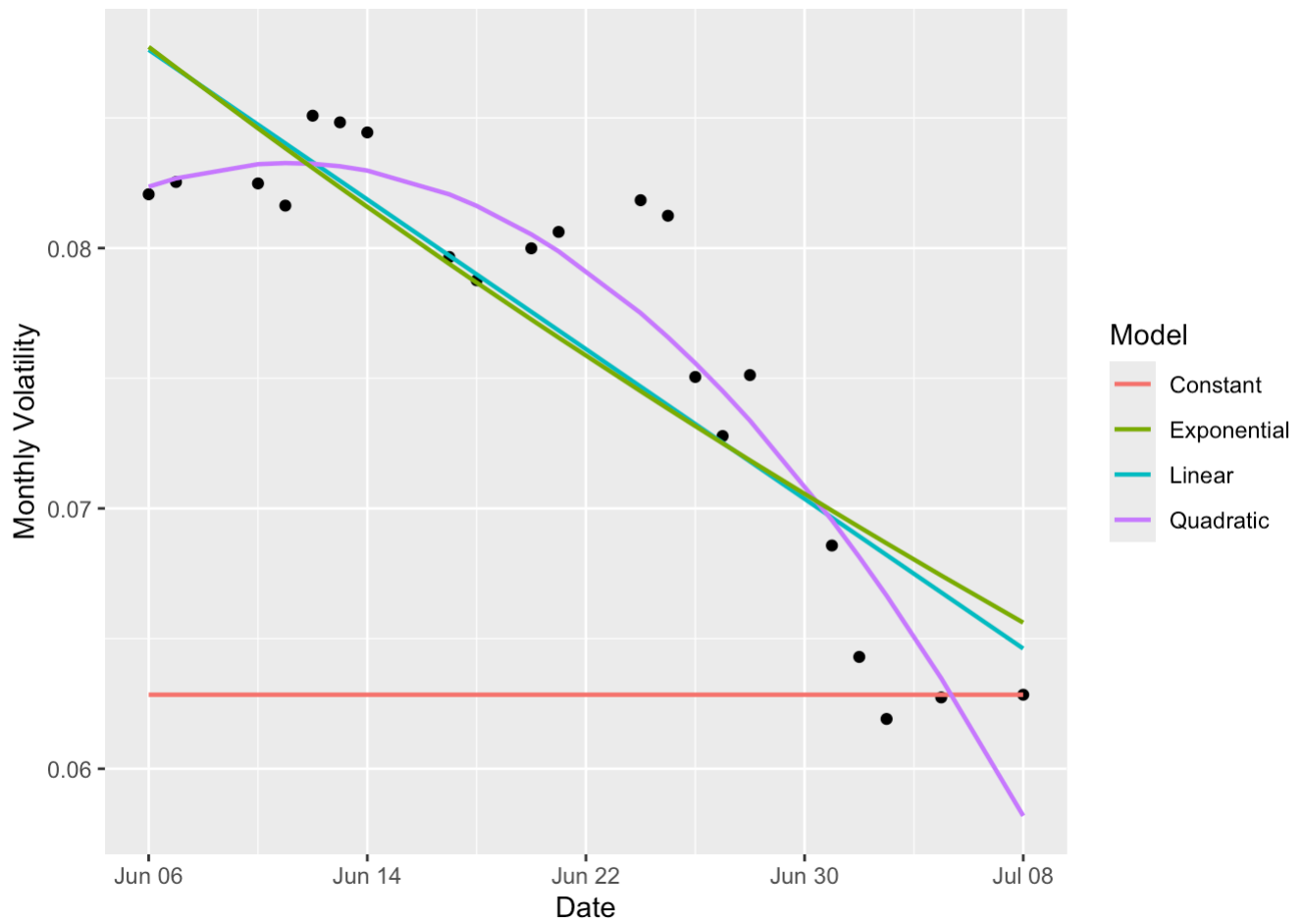
```
plot_vol("2024-07-01")
```



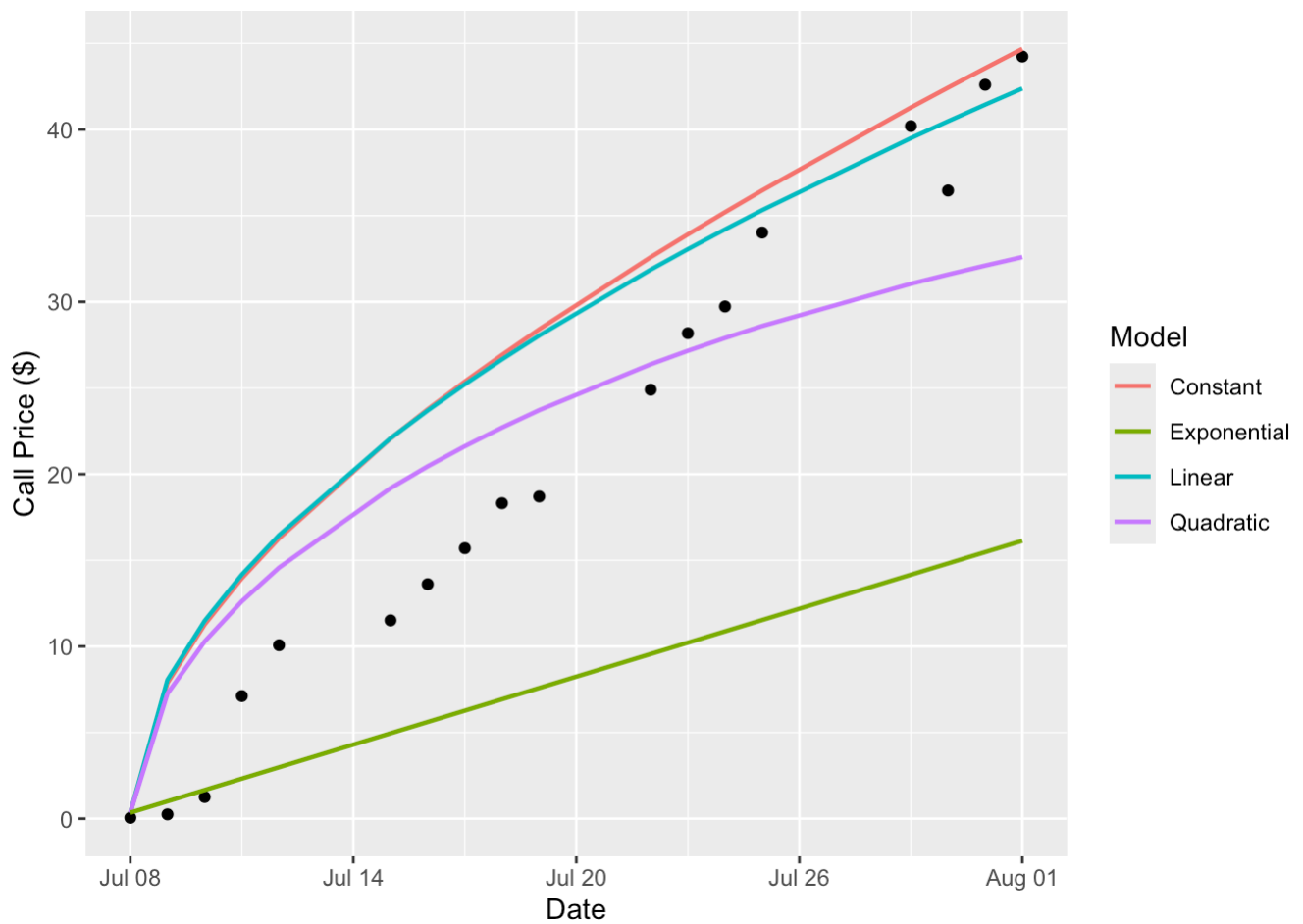
```
plot_call_prices("2024-07-01")
```



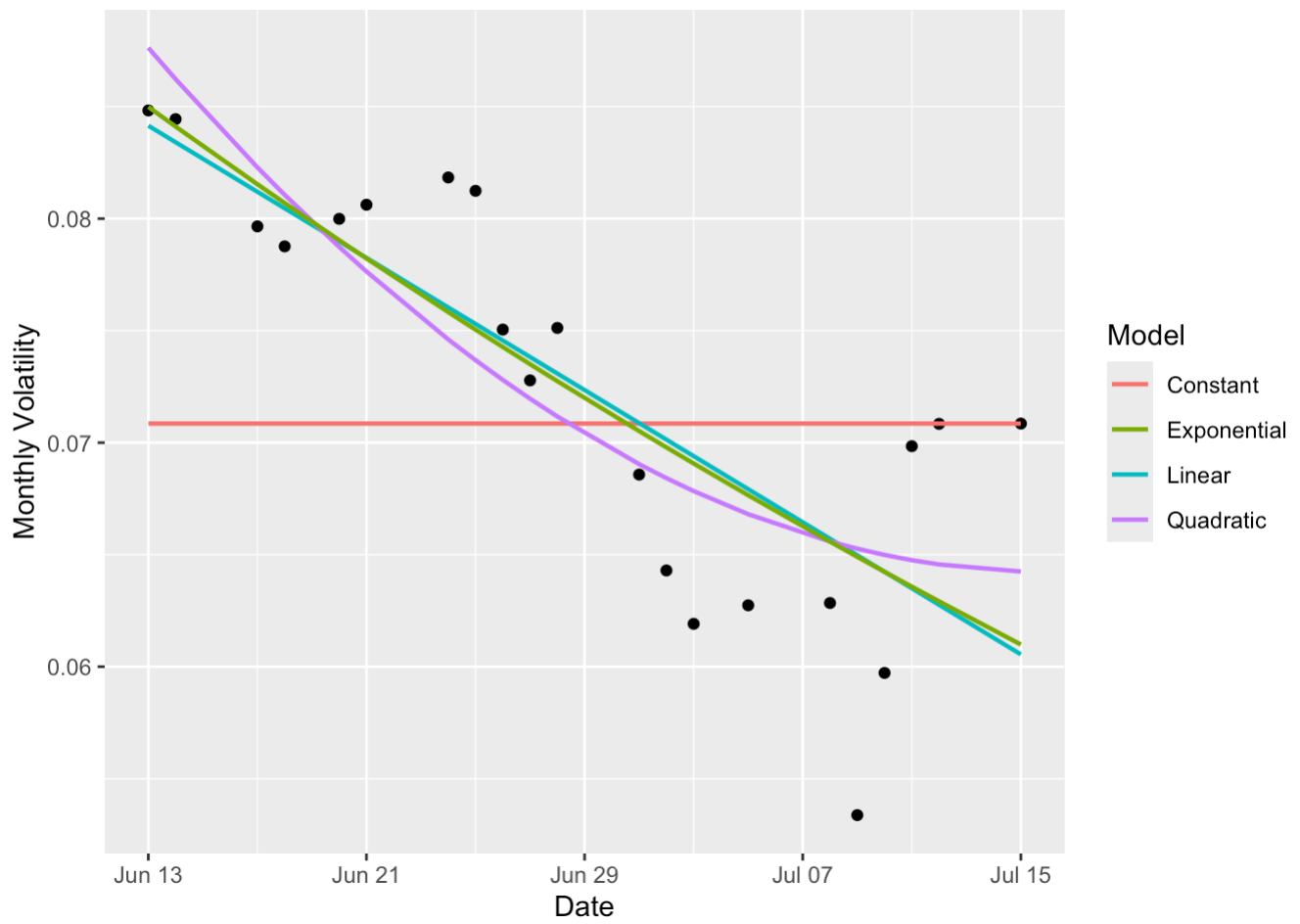
```
plot_vol("2024-07-08")
```



```
plot_call_prices("2024-07-08")
```

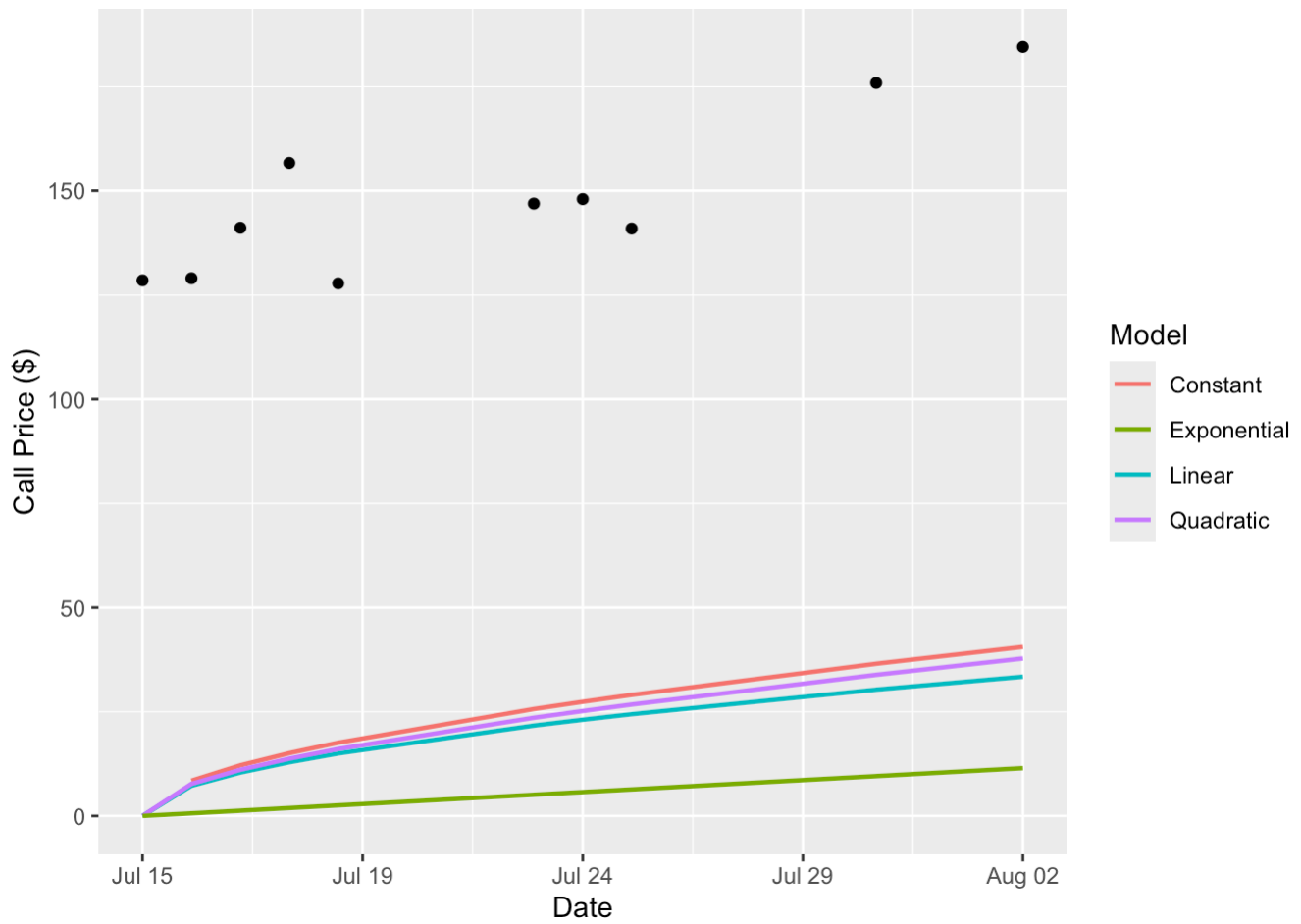


```
plot_vol("2024-07-15")
```

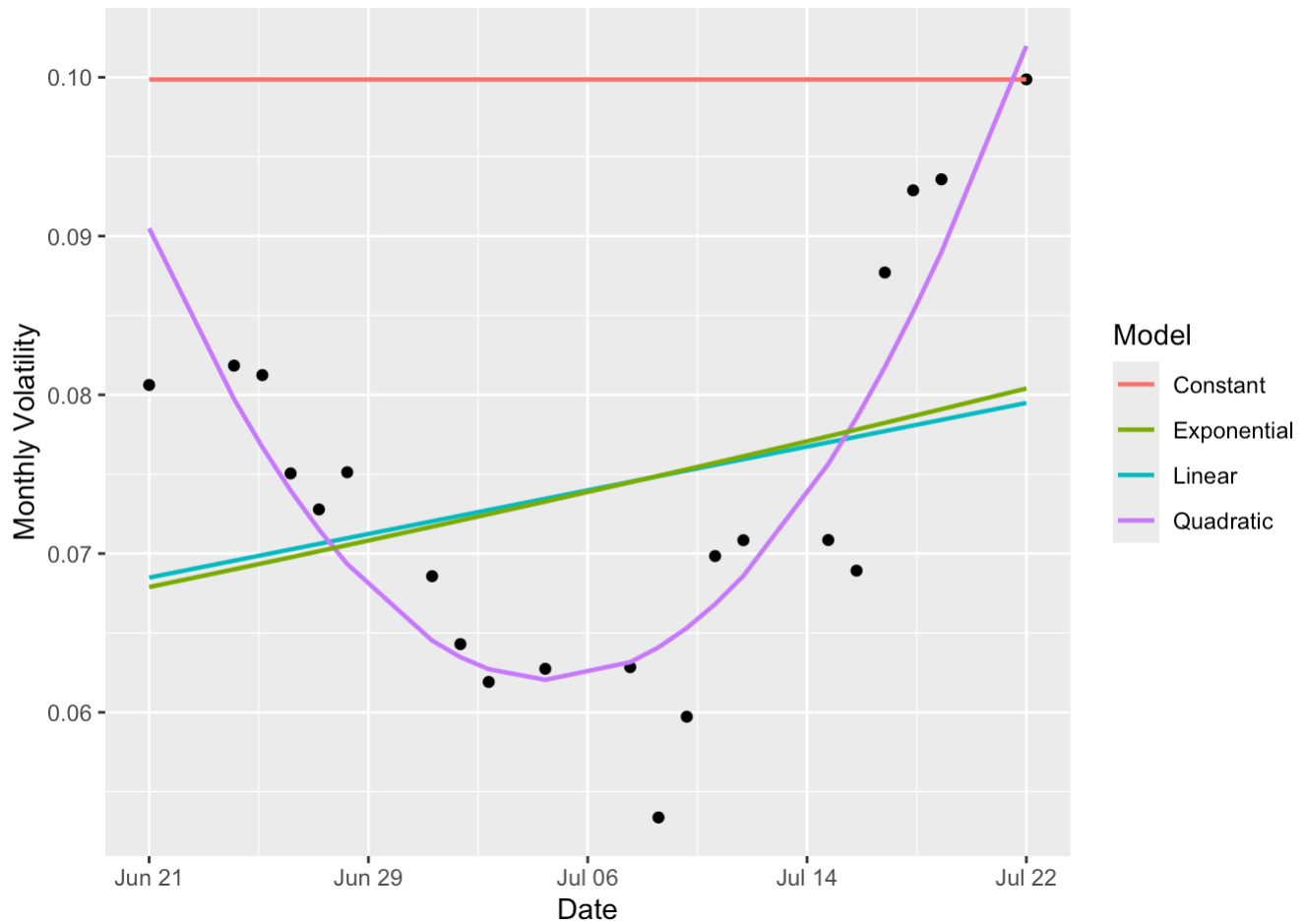


```
plot_call_prices("2024-07-15")
```

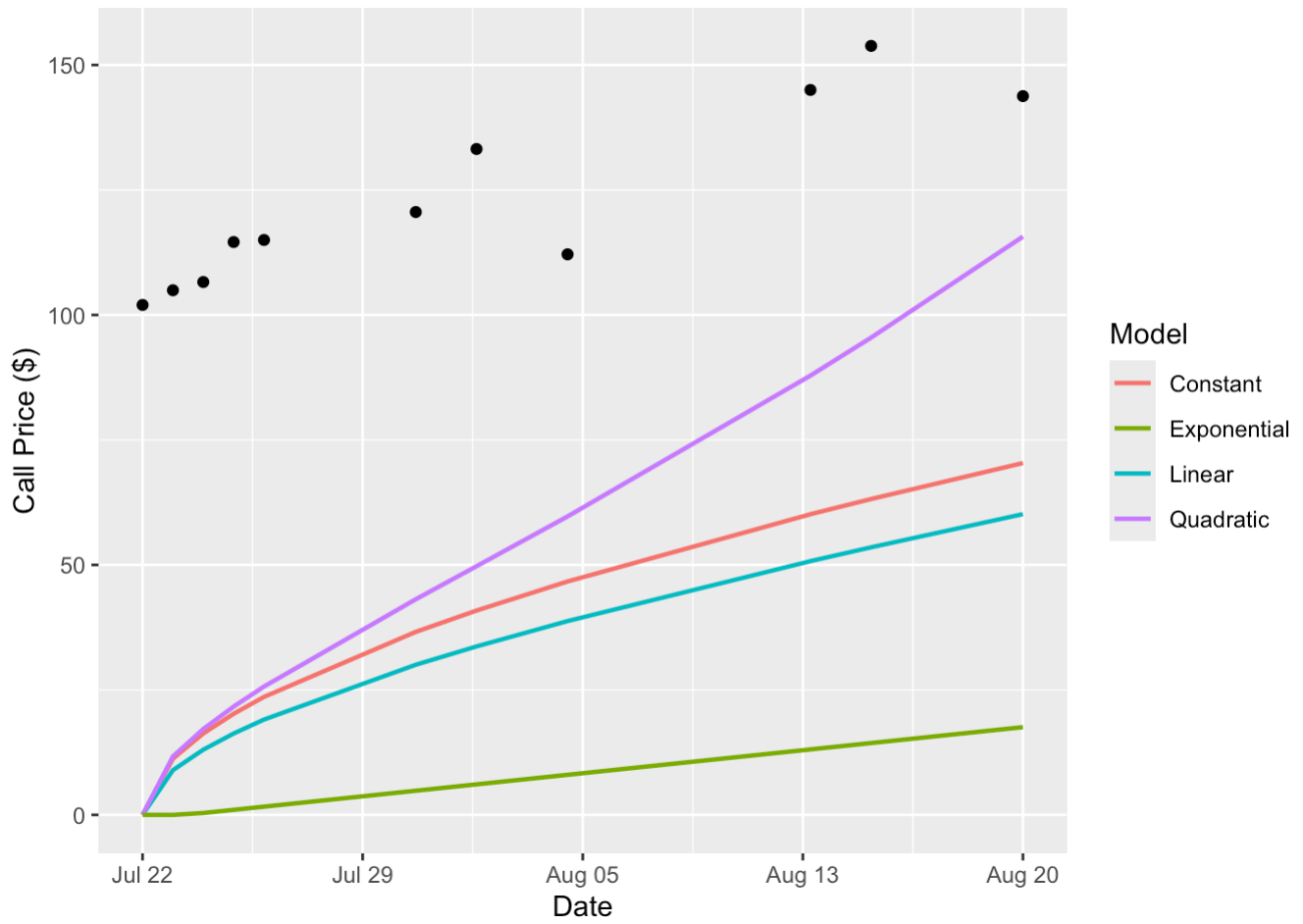
```
## Warning: Removed 1 row containing missing values or values outside the scale range  
## (`geom_line()`).
```



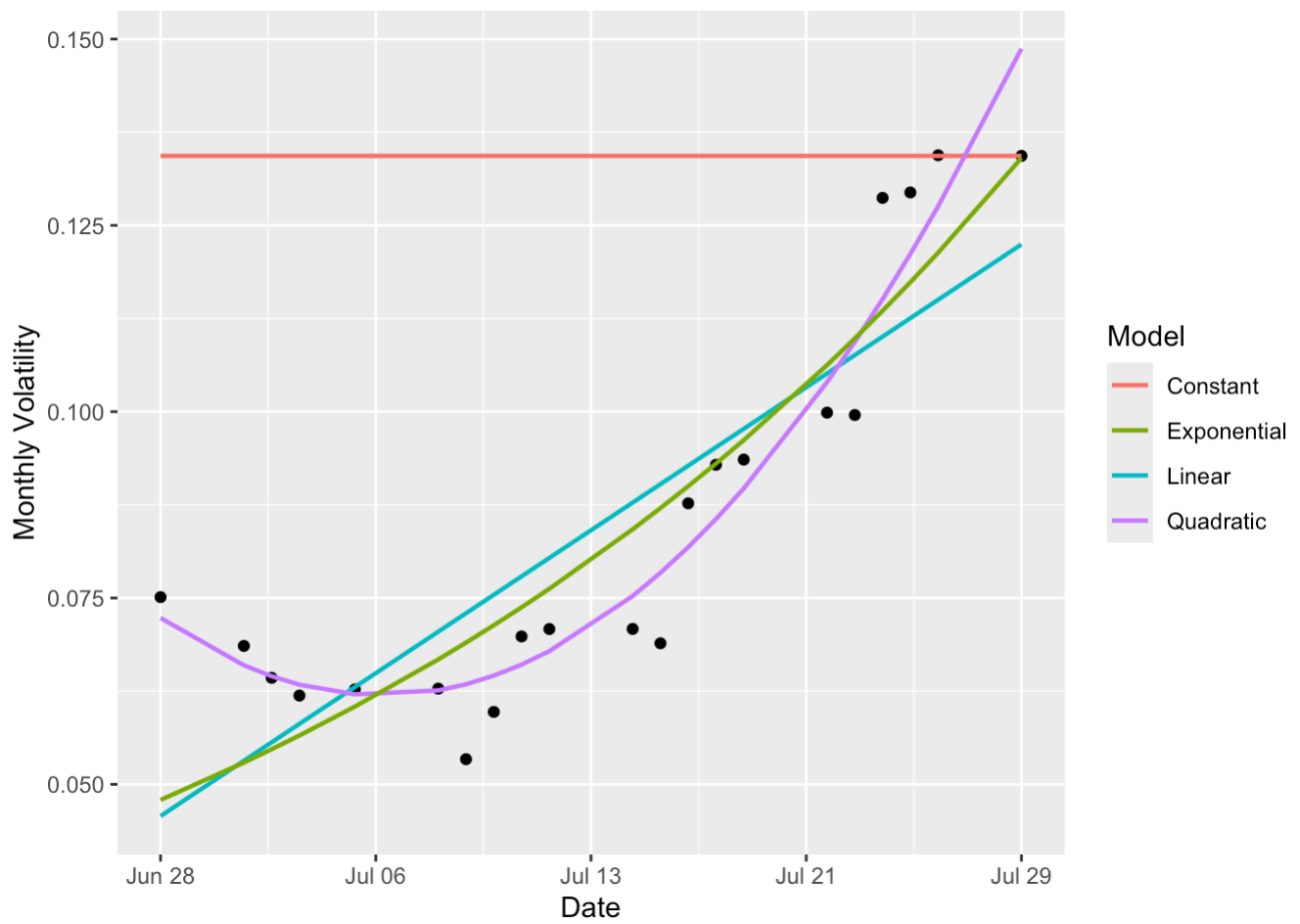
```
plot_vol("2024-07-22")
```



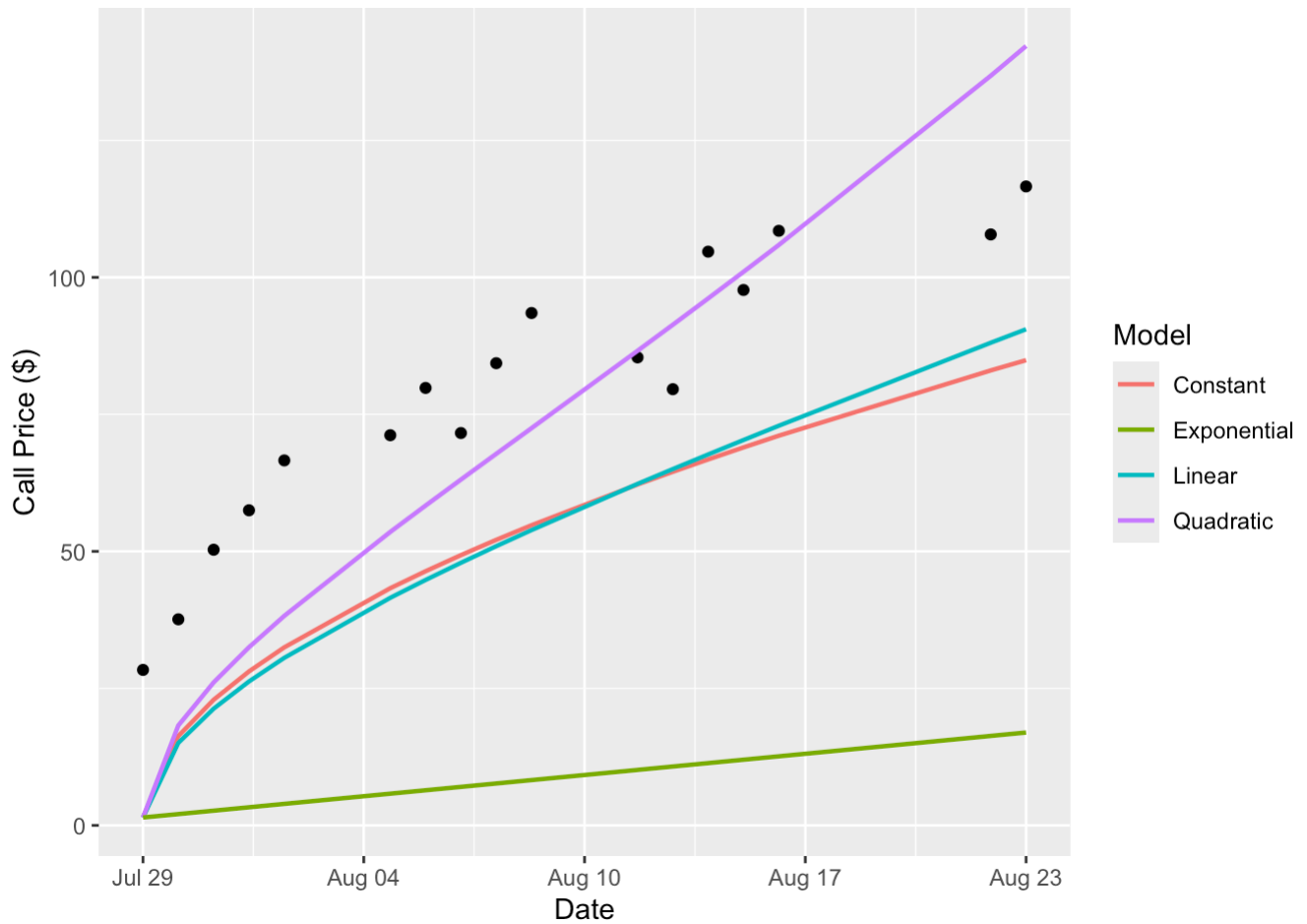

```
plot_call_prices("2024-07-22")
```



```
plot_vol("2024-07-29")
```



```
plot_call_prices("2024-07-29")
```



Output model performance for call prices on all dates

```
errs_tbl <- function(obs_date){
  vol_dat  <- hist_vol(obs_date) %>% fit_vol()
  calls_dat <- call_tbl_fun(obs_date, vol_dat) %>%
    select(-c(1:2), -c(5:13))
  return(calls_dat)
}
```

```
errs_tbl("2024-05-27")
```

```
## # A tibble: 17 × 10
##   `Obs. Date` `Exp. Date` err_const err_lin err_quad err_exp abs_err_const
##   <date>      <date>      <dbl>  <dbl>  <dbl>  <dbl>      <dbl>
## 1 2024-05-27 2024-05-28      15.4   15.5   17.8   15.3      15.4
## 2 2024-05-27 2024-05-29      12.9   13.1   16.6   17.7      12.9
## 3 2024-05-27 2024-05-30      16.9   17.2   21.7   25.4      16.9
## 4 2024-05-27 2024-05-31      18.3   18.8   24.2   30.0      18.3
## 5 2024-05-27 2024-06-03      11.1   12.2   20.4   30.5      11.1
## 6 2024-05-27 2024-06-04      13.9   15.1   24.2   35.4      13.9
## 7 2024-05-27 2024-06-05      15.6   17.0   27.1   39.1      15.6
## 8 2024-05-27 2024-06-07      15.0   16.9   29.0   42.2      15.0
## 9 2024-05-27 2024-06-10      12.1   14.7   29.8   44.2      12.1
## 10 2024-05-27 2024-06-11       5.84   8.71   24.9   39.5       5.84
## 11 2024-05-27 2024-06-12       5.45   8.58   25.8   40.6       5.45
## 12 2024-05-27 2024-06-14      12.1   15.8   35.1   50.0      12.1
## 13 2024-05-27 2024-06-17       1.88   6.46   28.8   43.6       1.88
## 14 2024-05-27 2024-06-20       7.13   12.6   37.6   52.3       7.13
## 15 2024-05-27 2024-06-21      18.9   24.8   50.5   65.2      18.9
## 16 2024-05-27 2024-06-26       8.73   16.3   44.5   60.1       8.73
## 17 2024-05-27 2024-06-27      14.9   22.8   51.2   67.2      14.9
## # i 3 more variables: abs_err_lin <dbl>, abs_err_quad <dbl>, abs_err_exp <dbl>
```

```
errs_tbl("2024-06-03")
```

```
## # A tibble: 18 × 10
##   `Obs. Date` `Exp. Date` err_const err_lin err_quad err_exp abs_err_const
##   <date>      <date>      <dbl>  <dbl>   <dbl>   <dbl>      <dbl>
## 1 2024-06-03 2024-06-03    -1.94  -1.94   -1.94  -1.94      1.94
## 2 2024-06-03 2024-06-04   -11.3  -11.2   -11.8  -2.37     11.3
## 3 2024-06-03 2024-06-05   -14.6  -14.4   -15.2  -1.80     14.6
## 4 2024-06-03 2024-06-06   -16.6  -16.3   -17.4  -0.917    16.6
## 5 2024-06-03 2024-06-07   -15.0  -14.5   -15.8   3.10     15.0
## 6 2024-06-03 2024-06-10   -20.8  -19.6   -21.6   3.01     20.8
## 7 2024-06-03 2024-06-11   -21.2  -19.7   -22.0   4.15     21.2
## 8 2024-06-03 2024-06-12   -15.0  -13.2   -15.7  11.8     15.0
## 9 2024-06-03 2024-06-13   -17.6  -15.5   -18.3  10.5     17.6
## 10 2024-06-03 2024-06-14   -19.5  -17.1   -20.2   9.91     19.5
## 11 2024-06-03 2024-06-17   -24.1  -20.7   -24.7   8.66     24.1
## 12 2024-06-03 2024-06-18   -18.1  -14.4   -18.7  15.7     18.1
## 13 2024-06-03 2024-06-20   -22.1  -17.6   -22.5  13.7     22.1
## 14 2024-06-03 2024-06-21   -20.0  -15.1   -20.4  16.6     20.0
## 15 2024-06-03 2024-06-24   -16.6  -10.5   -16.9  22.6     16.6
## 16 2024-06-03 2024-06-25   -31.0  -24.4   -31.2   8.99     31.0
## 17 2024-06-03 2024-06-27   -25.0  -17.5   -25.1  16.6     25.0
## 18 2024-06-03 2024-06-28   -27.0  -19.1   -27.0  15.3     27.0
## # i 3 more variables: abs_err_lin <dbl>, abs_err_quad <dbl>, abs_err_exp <dbl>
```

```
errs_tbl("2024-06-10")
```

```
## # A tibble: 21 × 10
##   `Obs. Date` `Exp. Date` err_const err_lin err_quad err_exp abs_err_const
##   <date>      <date>      <dbl>  <dbl>   <dbl>   <dbl>      <dbl>
## 1 2024-06-10 2024-06-10    -1.55  -1.55   -1.55  -1.55      1.55
## 2 2024-06-10 2024-06-11   -10.5  -10.1   -10.2  -2.21     10.5
## 3 2024-06-10 2024-06-12   -11.1  -10.5   -10.7   0.670     11.1
## 4 2024-06-10 2024-06-13   -11.0  -10.1   -10.4   3.51     11.0
## 5 2024-06-10 2024-06-14   -12.1  -11.0   -11.3   4.54     12.1
## 6 2024-06-10 2024-06-17   -17.6  -15.6   -16.1   4.15     17.6
## 7 2024-06-10 2024-06-18   -18.1  -15.9   -16.4   5.00     18.1
## 8 2024-06-10 2024-06-20   -21.6  -18.6   -19.3   4.05     21.6
## 9 2024-06-10 2024-06-21   -20.9  -17.6   -18.4   5.84     20.9
## 10 2024-06-10 2024-06-24   -22.3  -17.9   -18.9   7.50     22.3
## # i 11 more rows
## # i 3 more variables: abs_err_lin <dbl>, abs_err_quad <dbl>, abs_err_exp <dbl>
```

```
errs_tbl("2024-06-17")
```

```
## # A tibble: 18 × 10
##   `Obs. Date` `Exp. Date` err_const err_lin err_quad err_exp abs_err_const
##   <date>      <date>      <dbl>  <dbl>  <dbl>  <dbl>      <dbl>
## 1 2024-06-17 2024-06-17      8.12   8.12   8.12   8.12      8.12
## 2 2024-06-17 2024-06-18      8.08   8.52   8.24  17.0      8.08
## 3 2024-06-17 2024-06-20      6.45   7.38   6.82  21.4      6.45
## 4 2024-06-17 2024-06-21      9.01  10.2   9.51  26.1      9.01
## 5 2024-06-17 2024-06-24      7.49   9.43   8.37  29.5      7.49
## 6 2024-06-17 2024-06-25     17.3  19.5   18.3  40.7     17.3
## 7 2024-06-17 2024-06-26      9.04  11.5   10.2  33.7      9.04
## 8 2024-06-17 2024-06-27     12.1  14.8   13.4  37.9     12.1
## 9 2024-06-17 2024-06-28     13.1  16.2   14.6  40.0     13.1
##10 2024-06-17 2024-07-01     18.5  22.5   20.4  48.4     18.5
##11 2024-06-17 2024-07-02     14.1  18.4   16.1  44.8     14.1
##12 2024-06-17 2024-07-03     20.7  25.4   22.9  52.3     20.7
##13 2024-06-17 2024-07-05     24.0  29.3   26.5  57.2     24.0
##14 2024-06-17 2024-07-08     14.3  20.6   17.3  49.7     14.3
##15 2024-06-17 2024-07-09     23.8  30.5   26.9  59.8     23.8
##16 2024-06-17 2024-07-10     20.4  27.5   23.7  57.2     20.4
##17 2024-06-17 2024-07-12     25.7  33.5   29.3  63.7     25.7
##18 2024-06-17 2024-07-17     23.9  33.7   28.3  64.7     23.9
## # i 3 more variables: abs_err_lin <dbl>, abs_err_quad <dbl>, abs_err_exp <dbl>
```

```
errs_tbl("2024-06-24")
```

```
## # A tibble: 22 × 10
##   `Obs. Date` `Exp. Date` err_const err_lin err_quad err_exp abs_err_const
##   <date>      <date>      <dbl>  <dbl>  <dbl>  <dbl>      <dbl>
## 1 2024-06-24 2024-06-24    -0.430 -0.430   -0.430 -0.430      0.430
## 2 2024-06-24 2024-06-25    -2.80  -1.81   -2.73   5.98      2.80
## 3 2024-06-24 2024-06-26    -1.46  -0.00752 -1.38  10.9      1.46
## 4 2024-06-24 2024-06-27    -0.581  1.26   -0.516  14.4      0.581
## 5 2024-06-24 2024-06-28      3.34  5.55      3.38  20.5      3.34
## 6 2024-06-24 2024-07-01    -0.320  2.89   -0.462  22.0      0.320
## 7 2024-06-24 2024-07-02      8.54  12.1      8.32  32.2      8.54
## 8 2024-06-24 2024-07-03     10.9  14.8     10.6  35.9     10.9
## 9 2024-06-24 2024-07-05      2.59  7.12      2.03  29.9      2.59
##10 2024-06-24 2024-07-08      8.73  14.2      7.71  39.1      8.73
## # i 12 more rows
## # i 3 more variables: abs_err_lin <dbl>, abs_err_quad <dbl>, abs_err_exp <dbl>
```

```
errs_tbl("2024-07-01")
```

```
## # A tibble: 15 × 10
##   `Obs. Date` `Exp. Date` err_const err_lin err_quad err_exp abs_err_const
##   <date>      <date>      <dbl>  <dbl>   <dbl>  <dbl>      <dbl>
## 1 2024-07-01 2024-07-01    -2.14  -2.14   -2.14 -2.14      2.14
## 2 2024-07-01 2024-07-02    -9.44  -9.94   -9.90 -2.82      9.44
## 3 2024-07-01 2024-07-03   -13.1  -13.7   -13.7 -3.48     13.1
## 4 2024-07-01 2024-07-05   -18.1  -19.0   -18.9 -4.54     18.1
## 5 2024-07-01 2024-07-08   -23.6  -24.6   -24.5 -5.87     23.6
## 6 2024-07-01 2024-07-09   -24.3  -25.3   -25.1 -5.35     24.3
## 7 2024-07-01 2024-07-10   -24.7  -25.7   -25.5 -4.72     24.7
## 8 2024-07-01 2024-07-11   -24.3  -25.3   -25.0 -3.36     24.3
## 9 2024-07-01 2024-07-12   -22.5  -23.5   -23.2 -0.698     22.5
##10 2024-07-01 2024-07-15   -27.6  -28.4   -28.1 -3.28     27.6
##11 2024-07-01 2024-07-16   -26.8  -27.6   -27.2 -1.78     26.8
##12 2024-07-01 2024-07-17   -27.4  -28.1   -27.7 -1.70     27.4
##13 2024-07-01 2024-07-19   -27.1  -27.7   -27.2 -0.0813    27.1
##14 2024-07-01 2024-07-26   -24.3  -24.2   -23.5  6.64     24.3
##15 2024-07-01 2024-07-31   -25.3  -24.7   -23.8  7.86     25.3
## # i 3 more variables: abs_err_lin <dbl>, abs_err_quad <dbl>, abs_err_exp <dbl>
```

```
errs_tbl("2024-07-08")
```

```
## # A tibble: 18 × 10
##   `Obs. Date` `Exp. Date` err_const err_lin err_quad err_exp abs_err_const
##   <date>      <date>      <dbl>  <dbl>   <dbl>  <dbl>      <dbl>
## 1 2024-07-08 2024-07-08    -0.300 -0.300   -0.300 -0.300      0.300
## 2 2024-07-08 2024-07-09    -7.63  -7.81   -7.01 -0.758      7.63
## 3 2024-07-08 2024-07-10   -10.0  -10.2   -9.03 -0.407     10.0
## 4 2024-07-08 2024-07-11    -6.84  -7.05   -5.50  4.80      6.84
## 5 2024-07-08 2024-07-12    -6.20  -6.39   -4.50  7.09      6.20
## 6 2024-07-08 2024-07-15   -10.6  -10.6   -7.67  6.55     10.6
## 7 2024-07-08 2024-07-16   -10.1  -10.1   -6.84  8.00     10.1
## 8 2024-07-08 2024-07-17    -9.67  -9.52   -5.92  9.43      9.67
## 9 2024-07-08 2024-07-18    -8.61  -8.35   -4.39 11.4      8.61
##10 2024-07-08 2024-07-19    -9.70  -9.34   -5.01 11.1      9.70
##11 2024-07-08 2024-07-22    -7.69  -6.97   -1.48 15.3      7.69
##12 2024-07-08 2024-07-23    -5.73  -4.88    1.02 18.0      5.73
##13 2024-07-08 2024-07-24    -5.47  -4.48    1.83 18.9      5.47
##14 2024-07-08 2024-07-25    -2.44  -1.30    5.42 22.5      2.44
##15 2024-07-08 2024-07-29    -1.08  0.700    9.15 26.0      1.08
##16 2024-07-08 2024-07-30    -5.97  -4.02    4.87 21.6      5.97
##17 2024-07-08 2024-07-31   -0.966  1.16   10.5 27.1      0.966
##18 2024-07-08 2024-08-01   -0.445  1.86   11.6 28.1      0.445
## # i 3 more variables: abs_err_lin <dbl>, abs_err_quad <dbl>, abs_err_exp <dbl>
```

```
errs_tbl("2024-07-15")
```

```
## # A tibble: 10 × 10
##   `Obs. Date` `Exp. Date` err_const err_lin err_quad err_exp abs_err_const
##   <date>      <date>      <dbl>  <dbl>  <dbl>  <dbl>      <dbl>
## 1 2024-07-15 2024-07-15      NaN   129.   129.   129.      NaN
## 2 2024-07-15 2024-07-16    121.   122.   121.   128.    121.
## 3 2024-07-15 2024-07-17    129.   131.   130.   140.    129.
## 4 2024-07-15 2024-07-18    142.   144.   143.   155.    142.
## 5 2024-07-15 2024-07-19    110.   113.   112.   125.    110.
## 6 2024-07-15 2024-07-23    121.   125.   123.   142.    121.
## 7 2024-07-15 2024-07-24    121.   125.   123.   142.    121.
## 8 2024-07-15 2024-07-25    112.   117.   114.   135.    112.
## 9 2024-07-15 2024-07-30    139.   146.   142.   166.    139.
## 10 2024-07-15 2024-08-02    144.   151.   147.   173.    144.
## # i 3 more variables: abs_err_lin <dbl>, abs_err_quad <dbl>, abs_err_exp <dbl>
```

```
errs_tbl("2024-07-22")
```

```
## # A tibble: 11 × 10
##   `Obs. Date` `Exp. Date` err_const err_lin err_quad err_exp abs_err_const
##   <date>      <date>      <dbl>  <dbl>  <dbl>  <dbl>      <dbl>
## 1 2024-07-22 2024-07-22    102    102    102    102    102
## 2 2024-07-22 2024-07-23    93.7   96.0   93.3   105.    93.7
## 3 2024-07-22 2024-07-24    90.3   93.5   89.4   106.    90.3
## 4 2024-07-22 2024-07-25    94.4   98.3   92.9   114.    94.4
## 5 2024-07-22 2024-07-26    91.4   95.9   89.3   113.    91.4
## 6 2024-07-22 2024-07-31    84.0   90.6   77.5   116.    84.0
## 7 2024-07-22 2024-08-02    92.3   99.5   83.5   127.    92.3
## 8 2024-07-22 2024-08-05    65.4   73.4   52.4   104.    65.4
## 9 2024-07-22 2024-08-13    84.9   94.3   57.1   132.    84.9
## 10 2024-07-22 2024-08-15    90.6  100.    58.4   139.    90.6
## 11 2024-07-22 2024-08-20    73.4   83.6   28.1   126.    73.4
## # i 3 more variables: abs_err_lin <dbl>, abs_err_quad <dbl>, abs_err_exp <dbl>
```

```
errs_tbl("2024-07-29")
```

```
## # A tibble: 17 × 10
##   `Obs. Date` `Exp. Date` err_const err_lin err_quad err_exp abs_err_const
##   <date>      <date>      <dbl>  <dbl>  <dbl>  <dbl>      <dbl>
## 1 2024-07-29 2024-07-29      26.9   26.9   26.9   26.9      26.9
## 2 2024-07-29 2024-07-30      21.3   22.6   19.4   35.5      21.3
## 3 2024-07-29 2024-07-31      27.4   29.0   24.2   47.6      27.4
## 4 2024-07-29 2024-08-01      29.4   31.2   25.0   54.2      29.4
## 5 2024-07-29 2024-08-02      34.1   36.0   28.4   62.7      34.1
## 6 2024-07-29 2024-08-05      27.9   29.7   17.6   65.4      27.9
## 7 2024-07-29 2024-08-06      33.4   35.0   21.5   73.4      33.4
## 8 2024-07-29 2024-08-07      22.3   23.7    8.50   64.6      22.3
## 9 2024-07-29 2024-08-08      32.2   33.4   16.5   76.7      32.2
##10 2024-07-29 2024-08-09      38.7   39.6   21.0   85.2      38.7
##11 2024-07-29 2024-08-12      23.2   23.1   -1.20   75.3      23.2
##12 2024-07-29 2024-08-13      15.1   14.6  -11.8   68.9      15.1
##13 2024-07-29 2024-08-14      37.9   37.0    8.54   93.3      37.9
##14 2024-07-29 2024-08-15      28.7   27.4   -3.31   85.7      28.7
##15 2024-07-29 2024-08-16      37.4   35.6    2.58   95.9      37.4
##16 2024-07-29 2024-08-22      24.8   19.8  -29.0   91.5      24.8
##17 2024-07-29 2024-08-23      31.7   26.1  -25.6   99.7      31.7
## # i 3 more variables: abs_err_lin <dbl>, abs_err_quad <dbl>, abs_err_exp <dbl>
```

Summarize model performance for call prices on all dates

```
summary_tbl <- map_dfr(obs_dates, function(d) {
  errs_tbl(d) %>%
    summarise(
      Date      = d,
      RMSE_const = sqrt(mean(err_const^2, na.rm = TRUE)),
      RMSE_lin   = sqrt(mean(err_lin^2,   na.rm = TRUE)),
      RMSE_quad  = sqrt(mean(err_quad^2,  na.rm = TRUE)),
      RMSE_exp   = sqrt(mean(err_exp^2,   na.rm = TRUE)),
      MAE_const  = mean(abs_err_const, na.rm = TRUE),
      MAE_lin    = mean(abs_err_lin,   na.rm = TRUE),
      MAE_quad   = mean(abs_err_quad,  na.rm = TRUE),
      MAE_exp    = mean(abs_err_exp,   na.rm = TRUE)
    )
}) %>%
  bind_rows(
    summarise(.,
      Date      = "Mean",
      RMSE_const = mean(RMSE_const),
      RMSE_lin   = mean(RMSE_lin),
      RMSE_quad  = mean(RMSE_quad),
      RMSE_exp   = mean(RMSE_exp),
      MAE_const  = mean(MAE_const),
      MAE_lin    = mean(MAE_lin),
      MAE_quad   = mean(MAE_quad),
      MAE_exp    = mean(MAE_exp)
    )
  )
summary_tbl
```



```
## # A tibble: 11 × 9
##   Date      RMSE_const RMSE_lin RMSE_quad RMSE_exp MAE_const MAE_lin MAE_quad
##   <chr>      <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
## 1 2024-05-27    13.0     15.8     31.6     43.6     12.1     15.1     29.9
## 2 2024-06-03    19.8     16.4     20.2     11.3     18.7     15.7     19.2
## 3 2024-06-10    20.4     14.9     16.0      9.83     19.3     14.3     15.4
## 4 2024-06-17    16.6     21.2     18.7     44.8     15.3     19.3     17.2
## 5 2024-06-24    13.9     20.1     11.9     43.4     11.1     16.6      9.23
## 6 2024-07-01    22.6     23.2     22.9      4.20     21.4     22.0     21.7
## 7 2024-07-08      7.02      6.80      6.48     16.0      6.08      5.83      5.67
## 8 2024-07-15   127.     131.     129.     144.     127.     130.     128.
## 9 2024-07-22    88.0     93.7     78.0     117.     87.5     93.4     74.9
## 10 2024-07-29    29.6     29.6     19.4     73.6     29.0     28.9     17.1
## 11 Mean         35.8     37.2     35.4     50.8     34.7     36.1     33.9
## # i 1 more variable: MAE_exp <dbl>
```

Best-fit model per date for call prices

```
best_fit_tbl <- summary_tbl %>%
  rowwise() %>%
  mutate(
    RMSE_Best_Fit = c("Constant", "Linear", "Quadratic", "Exponential")[
      which.min(c(RMSE_const, RMSE_lin, RMSE_quad, RMSE_exp))
    ],
    RMSE = min(RMSE_const, RMSE_lin, RMSE_quad, RMSE_exp) %>% round(2),

    MAE_Best_Fit = c("Constant", "Linear", "Quadratic", "Exponential")[
      which.min(c(MAE_const, MAE_lin, MAE_quad, MAE_exp))
    ],
    MAE = min(MAE_const, MAE_lin, MAE_quad, MAE_exp) %>% round(2)
  ) %>%
  ungroup() %>%
  select(Date, RMSE_Best_Fit, RMSE, MAE_Best_Fit, MAE)

best_fit_tbl
```

```
## # A tibble: 11 × 5
##   Date      RMSE_Best_Fit  RMSE MAE_Best_Fit  MAE
##   <chr>      <chr>      <dbl> <chr>      <dbl>
## 1 2024-05-27 Constant    13.0 Constant    12.1
## 2 2024-06-03 Exponential 11.2 Exponential  9.31
## 3 2024-06-10 Exponential  9.83 Exponential  8.44
## 4 2024-06-17 Constant    16.6 Constant    15.3
## 5 2024-06-24 Quadratic   11.9 Quadratic    9.23
## 6 2024-07-01 Exponential  4.2 Exponential  3.62
## 7 2024-07-08 Quadratic    6.48 Quadratic    5.67
## 8 2024-07-15 Constant   127. Constant   127.
## 9 2024-07-22 Quadratic   78.0 Quadratic   74.9
## 10 2024-07-29 Quadratic   19.4 Quadratic   17.1
## 11 Mean      Quadratic   35.4 Quadratic   33.9
```

Paired t-tests: volatility model errors vs. constant baseline

```
# Filter out the "Mean" summary row – keep only the 10 date rows
vol_test_dat <- vol_summary_tbl %>% filter(Date != "Mean")

# RMSE tests (volatility)
vol_rmse_lin <- t.test(vol_test_dat$RMSE_lin, vol_test_dat$RMSE_const, paired = TRUE)
vol_rmse_quad <- t.test(vol_test_dat$RMSE_quad, vol_test_dat$RMSE_const, paired = TRUE)
vol_rmse_exp <- t.test(vol_test_dat$RMSE_exp, vol_test_dat$RMSE_const, paired = TRUE)

# MAE tests (volatility)
vol_mae_lin <- t.test(vol_test_dat$MAE_lin, vol_test_dat$MAE_const, paired = TRUE)
vol_mae_quad <- t.test(vol_test_dat$MAE_quad, vol_test_dat$MAE_const, paired = TRUE)
vol_mae_exp <- t.test(vol_test_dat$MAE_exp, vol_test_dat$MAE_const, paired = TRUE)

# Compile results into a readable table
vol_ttest_results <- tibble(
  Model = rep(c("Linear", "Quadratic", "Exponential"), 2),
  Metric = rep(c("RMSE", "MAE"), each = 3),
  Mean_Diff = c(
    vol_rmse_lin$estimate, vol_rmse_quad$estimate, vol_rmse_exp$estimate,
    vol_mae_lin$estimate, vol_mae_quad$estimate, vol_mae_exp$estimate
  ),
  t_stat = c(
    vol_rmse_lin$statistic, vol_rmse_quad$statistic, vol_rmse_exp$statistic,
    vol_mae_lin$statistic, vol_mae_quad$statistic, vol_mae_exp$statistic
  ),
  p_value = c(
    vol_rmse_lin$p.value, vol_rmse_quad$p.value, vol_rmse_exp$p.value,
    vol_mae_lin$p.value, vol_mae_quad$p.value, vol_mae_exp$p.value
  ),
  CI_lower = c(
    vol_rmse_lin$conf.int[1], vol_rmse_quad$conf.int[1], vol_rmse_exp$conf.int[1],
    vol_mae_lin$conf.int[1], vol_mae_quad$conf.int[1], vol_mae_exp$conf.int[1]
  ),
  CI_upper = c(
    vol_rmse_lin$conf.int[2], vol_rmse_quad$conf.int[2], vol_rmse_exp$conf.int[2],
    vol_mae_lin$conf.int[2], vol_mae_quad$conf.int[2], vol_mae_exp$conf.int[2]
  )
) %>%
  mutate(across(where(is.numeric), \(x) round(x, 4)))

vol_ttest_results
```

```
## # A tibble: 6 × 7
##   Model      Metric Mean_Diff t_stat p_value CI_lower CI_upper
##   <chr>      <chr>      <dbl> <dbl> <dbl>   <dbl>   <dbl>
## 1 Linear      RMSE    -0.0173 -5.00  0.0007 -0.0251 -0.0095
## 2 Quadratic  RMSE    -0.0193 -4.85  0.0009 -0.0283 -0.0103
## 3 Exponential RMSE    -0.0176 -4.81  0.001  -0.0259 -0.0093
## 4 Linear      MAE     -0.0148 -4.70  0.0011 -0.022  -0.0077
## 5 Quadratic  MAE     -0.0166 -4.57  0.0013 -0.0249 -0.0084
## 6 Exponential MAE     -0.0151 -4.50  0.0015 -0.0227 -0.0075
```

Paired t-tests: call pricing model errors vs. constant baseline

```

call_test_dat <- summary_tbl %>% filter(Date != "Mean")

# RMSE tests (call pricing)
call_rmse_lin <- t.test(call_test_dat$RMSE_lin, call_test_dat$RMSE_const, paired =
TRUE)
call_rmse_quad <- t.test(call_test_dat$RMSE_quad, call_test_dat$RMSE_const, paired =
TRUE)
call_rmse_exp <- t.test(call_test_dat$RMSE_exp, call_test_dat$RMSE_const, paired =
TRUE)

# MAE tests (call pricing)
call_mae_lin <- t.test(call_test_dat$MAE_lin, call_test_dat$MAE_const, paired =
TRUE)
call_mae_quad <- t.test(call_test_dat$MAE_quad, call_test_dat$MAE_const, paired =
TRUE)
call_mae_exp <- t.test(call_test_dat$MAE_exp, call_test_dat$MAE_const, paired =
TRUE)

call_ttest_results <- tibble(
  Model = rep(c("Linear", "Quadratic", "Exponential"), 2),
  Metric = rep(c("RMSE", "MAE"), each = 3),
  Mean_Diff = c(
    call_rmse_lin$estimate, call_rmse_quad$estimate, call_rmse_exp$estimate,
    call_mae_lin$estimate, call_mae_quad$estimate, call_mae_exp$estimate
  ),
  t_stat = c(
    call_rmse_lin$statistic, call_rmse_quad$statistic, call_rmse_exp$statistic,
    call_mae_lin$statistic, call_mae_quad$statistic, call_mae_exp$statistic
  ),
  p_value = c(
    call_rmse_lin$p.value, call_rmse_quad$p.value, call_rmse_exp$p.value,
    call_mae_lin$p.value, call_mae_quad$p.value, call_mae_exp$p.value
  ),
  CI_lower = c(
    call_rmse_lin$conf.int[1], call_rmse_quad$conf.int[1], call_rmse_exp$conf.int
[1],
    call_mae_lin$conf.int[1], call_mae_quad$conf.int[1], call_mae_exp$conf.int[1]
  ),
  CI_upper = c(
    call_rmse_lin$conf.int[2], call_rmse_quad$conf.int[2], call_rmse_exp$conf.int
[2],
    call_mae_lin$conf.int[2], call_mae_quad$conf.int[2], call_mae_exp$conf.int[2]
  )
) %>%
  mutate(across(where(is.numeric), \(x) round(x, 4)))

call_ttest_results

```

```
## # A tibble: 6 × 7
##   Model      Metric Mean_Diff t_stat p_value CI_lower CI_upper
##   <chr>      <chr>      <dbl> <dbl> <dbl>   <dbl> <dbl>
## 1 Linear      RMSE         1.45  1.19  0.265   -1.31  4.20
## 2 Quadratic   RMSE        -0.386 -0.152  0.883   -6.15  5.37
## 3 Exponential RMSE        15.0   2.25  0.0512 -0.0963 30.2
## 4 Linear      MAE         1.42  1.25  0.243   -1.15  4.00
## 5 Quadratic   MAE        -0.827 -0.311  0.763   -6.84  5.19
## 6 Exponential MAE        14.0   2.16  0.0592 -0.670 28.8
```

```
write.csv(call_ttest_results, "Call_T-Test.csv")
```